



A new dynamic model for cylindrical roller bearings with flexible rollers and bearing sliding investigation

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ABSTRACT

Cylindrical roller bearings are inevitably impacted by external moments or mounting error, which leads to uneven load distribution on the rollers and triggers deformation. Current methods are insufficient to simulate the deformation while ensuring solution accuracy. To address this, a new slicing approach is innovatively proposed in the paper, where springs are added between the neighboring slices to simulate the elastic deformation of the roller under external loads. Compared with the existing methods, the method presented in this paper has the best agreement with the finite element model. On this basis, a dynamic model for cylindrical roller bearing with roller deformation is further developed and verified experimentally. Finally, the sliding behavior inside the bearing under three typical conditions is investigated. A rich spectrum of frequencies emerges in the bearing contact load between the roller and raceway because of the roller deformation. These are all integer multiples of the roller passage frequency. An interesting phenomenon is observed that the sliding velocity is strongly influenced by the orbital speed of the roller compared to its rotational speed. The modeling and analysis in this paper provide new directions to the future work.

1. Introduction

Cylindrical roller bearings are extensively used in aviation engines, new energy motors, gearboxes and other fields for their excellent load carrying and high speeds performance [1,2]. However, with the increasingly harsh conditions, they are exposed to more and more failures, such as excessive wear caused by severe sliding and bias loads [3,4]. With accurate numerical models, the bearing performance can be improved in the design phase to avoid such failures, thus reducing the development cycle and falling the cost of mechanical systems.

The cylindrical roller bearing model has been explored by many scholars for a long time. And a number of influential publications has emerged. Harris [5] firstly presented the numerical model for cylindrical roller bearing, named static model, to investigate the effect of misalignment on its fatigue life. On this basis, Liu [6] and Ramazan [7] introduced the driving speed to develop the quasi-static model and the effect of speed on its life was further studied. Since the role of the lubricant was ignored by them, the performance of cylindrical roller bearings could only be roughly predicted. Harris [8] later used an empirical approach to consider the lubricant and further constructed a quasi-dynamic model to analyze the sliding inside the bearing. Regrettably, however, the nonlinear

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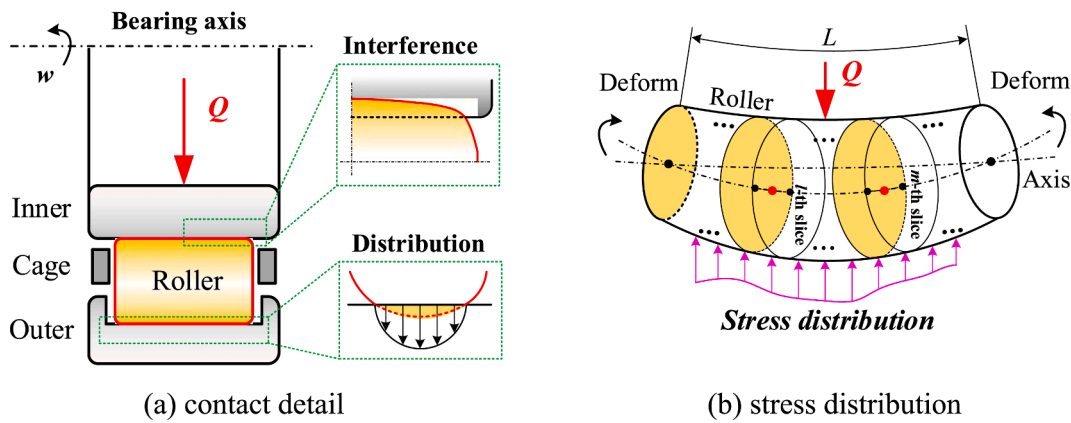


Fig. 1. Schematic of the contact between roller and raceway.

equations were used obtain steady-state values, making it unsuitable for its transient performance analysis [9].

To this end, Tu [10–12] have introduced nonlinear differential equations to construct a dynamic model for cylindrical roller bearings. The time-varying sliding feature inside the bearing under different conditions was further investigated. On this basis, Liu [13–16] studied the effects of surface roughness, raceway faults, etc. on the contact characteristics and vibration characteristics of bearings. Furthermore, Liu [17–19] made some improvements to the previous model with consideration of the cage flexibility and the lubricant-thermal interaction. The sliding performance inside the cylindrical roller bearing under accelerated conditions were further researched. However, the motion of the rollers was restricted in the radial plane by them, which means that the tilt and skew of the roller motion were ignored. The dynamic behavior of the bearing components therefore cannot be simulated by them.

Cao [20,21] used the slicing method (Slicing method of Hertzian contact) to model the complex motion of the roller, and further combined it with an empirical friction model to develop a dynamic model for cylindrical roller bearing. The shear stress at the roller/raceway interface were integrated by Han [22,23] to obtain their friction vector, and the sliding on this interface under time-varying load was further analyzed. Deng [24–28] derived a model for the resistance coefficient between the roller and raceway, the very famous five-parameter rheological model, through a large number of experiments. The effects of roller dynamic unbalance and lubricant characteristics on the cage behavior were further investigated. Although all of these models accounted for the complex motion of the roller using the slice method, the interactions between the slices were ignored. As Yang [29] pointed out, the interactions were essential for the predictions, such as “edge effect”. Later, he proposed an improved slicing method (Slicing method of influence coefficient) for obtaining the stress distribution at the roller/raceway interface. Due to the reliable accuracy of the solution, this method was also adopted by some scholars [30,31]. However, it has the disadvantage of assuming that the roller axis is not deformed, which is unrealistic for rollers subjected to heavy or biased loads. In such scenarios, rollers can experience skewed motion, leading to uneven stress distribution along the roller axis, deformation, and compromised bearing performance. Surface defects, such as wear or pitting, further complicate accurate predictions of dynamic responses.

To address this, several methods in finite element analysis (FEA) have been proposed to model cylindrical roller bearings with high accuracy, particularly for defective bearings [32–34]. For instance, Singh [32] proposed a finite element model of a defective bearing with LS-DYNA, and studied its contact forces and vibrations in both the time and frequency domains. The experiments were further conducted by him to validate the model [33]. Yang [34] also used the finite element method to construct a dynamic model of a cylindrical roller bearing with roller failures, and analyzed the impact of roller defects on cage slip. However, the high computational demands and the lack of lubrication consideration limit these models’ practical application in real-time engineering scenarios. As highlighted in recent works, combining analytical and numerical models can overcome these drawbacks [35,36]. Safian [37] has explored it by integrating lumped parameter models with FEA to enhance efficiency without sacrificing accuracy. However, these models lack real-time interaction between elements, raising concerns about their transient performance predictions.

For the cylindrical roller bearing dynamic model, the slice method is a good choice for simulating the complex motion of the roller due to the easy coupling with the numerical models. However, there are various problems with the current methods, which either ignore the interaction between slices or the roller elastic deformation. Therefore, based on the previous work [38,39], a slicing method of beam is innovatively proposed in this paper. The slice-slice interaction and the deformation are all well addressed. Compared with the two current slicing method, it results show a best agreement with finite element model. Combined with the existing dynamic model, a new dynamic model with roller flexibility for cylindrical roller bearings is firstly presented. The sliding behavior inside bearings under three typical operating conditions is further analyzed. The model’s framework can be extended to areas such as fault analysis and bearing failure reconstruction, offering potential applications in future research.

2. Discussion of different slicing methods

As shown in Fig. 1, the stress distribution on the roller surface is uneven because of its trimming and deformation. To simulate it, three different slicing methods are proposed in this section: Hertzian contact-based slicing method, influence coefficient-based slicing

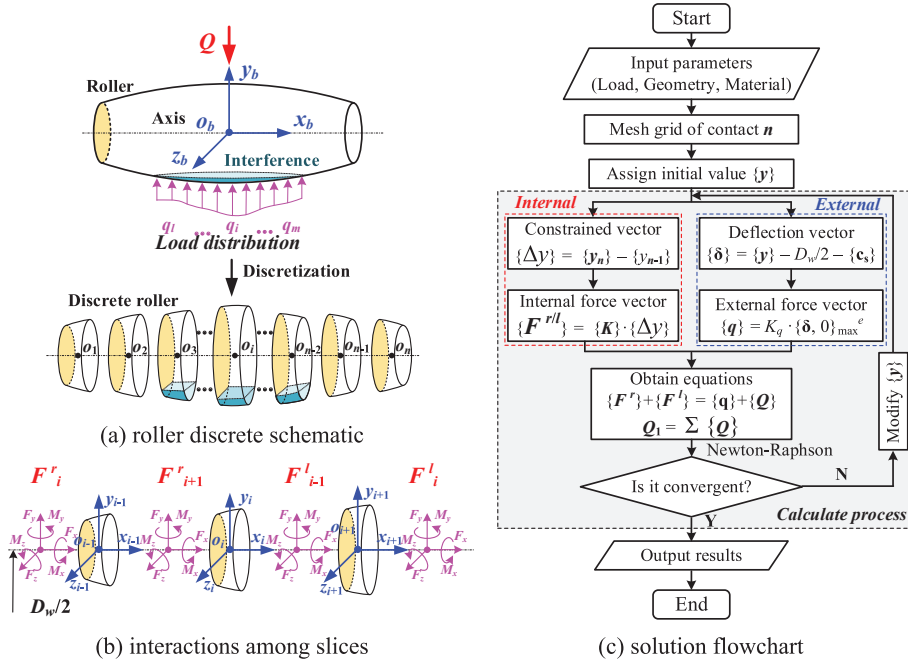


Fig. 2. Solution for roller stress distribution based on beam theory.

method and beam-based slicing method. The prediction results of these slicing method are further compared.

As mentioned in the introduction, the Hertzian contact-based slicing method divides the roller into n slices along its axis, calculates the load on each slice, and sums them to obtain the total external load on the roller. The influence coefficient-based method improves accuracy by accounting for slice interactions but overlooks the elastic deformation of the roller under load, which raises concerns about its precision. In this paper, a slicing method using beam theory is employed to discretize the roller and account for its elastic deformation, significantly enhancing the accuracy of previous models. Since the first two methods are more commonly, they are detailed in Appendix A and Appendix B, respectively. This section will focus on the modeling of the beam-based slicing method.

2.1. Slicing method based on beam

A new slicing method based on beam is presented innovatively in this section, in which each slice has independent degrees of freedom of motion. Its solution for stress distribution is displayed in Fig. 2. As shown in Fig. 2(a), the roller is first discretized into n slices of same width. For any slice, a contact load from the raceway is applied in addition to the constraint load from neighboring slices. Fig. 2(b) further gives the principle of this method. The spring and damping are added between the adject slices to simulate the roller deformation.

The essence of this method is to construct the relative displacements of the different slices, which is further covert into constraint vector based on beam theory. Therefore, it is necessary to define the displacement vector for each slice inside the roller. Taking the i -th slice as an example, its displacement vector can be expressed as:

$$X_i = [x_i \ y_i \ z_i \ \theta_{xi} \ \theta_{yi} \ \theta_{zi}]^T \quad (13)$$

As shown in Fig. 2(b), according to the Timoshenko beam theory, the constraint vector from the $(i-1)$ -th slice on the i -th slice can be defined by:

$$F_{i-1}^l = - \begin{bmatrix} K_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & K_{22} & 0 & 0 & 0 & K_{26} \\ 0 & 0 & K_{33} & 0 & K_{35} & 0 \\ 0 & 0 & 0 & K_{44} & 0 & 0 \\ 0 & 0 & K_{53} & 0 & K_{55} & 0 \\ 0 & K_{62} & 0 & 0 & 0 & K_{66} \end{bmatrix} \begin{bmatrix} x_i - x_{i-1} \\ y_i - y_{i-1} \\ z_i - z_{i-1} \\ \theta_{xi} - \theta_{xi-1} \\ \theta_{yi} - \theta_{yi-1} \\ \theta_{zi} - \theta_{zi-1} \end{bmatrix} - \begin{bmatrix} 2h \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (1)$$

where K is the deflection coefficient, a function of the structural and material coefficients [38,39], listed in Table 1.

where G is the shear modulus of the slice. I_{xx} is the torsional constant, and $I_{yy/zz}$ is the area moment of inertia. For the slice on the

Table 1
The deflection coefficients of Timoshenko beam.

Name	Formula	Name	Formula
K_{11}	$\frac{EA}{2h}$	K_{44}	$\frac{GI_{xx}}{2h}$
K_{22}	$\frac{12EI_{zz}}{(2h)^3(1+P_y)}$	K_{53}	$\frac{6EI_{yy}}{(2h)^2(1+P_z)}$
K_{26}	$\frac{-6EI_{zz}}{(2h)^3(1+P_y)}$	K_{55}	$\frac{(4+P_z)EI_{yy}}{2h(1+P_z)}$
K_{33}	$\frac{12EI_{yy}}{(2h)^3(1+P_z)}$	K_{26}	$\frac{-6EI_{zz}}{(2h)^3(1+P_y)}$
K_{35}	$\frac{6EI_{yy}}{(2h)^2(1+P_z)}$	K_{66}	$\frac{(4+P_y)EI_{zz}}{2h(1+P_y)}$

roller, they can be defined by:

$$\begin{cases} I_{xx} = 0.5\pi r^4 \\ I_{yy} = 0.25\pi r^4 \\ I_{zz} = 0.25\pi r^4 \end{cases} \quad (2)$$

where r is the diameter of the slice.

$P_{y/z}$ is a coefficient for the shear deflection of Timoshenko beams, which can be calculated by:

$$\begin{cases} P_y = \frac{12EI_{yy}}{GA(2h)^2I_z^2} \int_A \left(\frac{\tau_z}{I_y} \right)^2 dA \\ P_z = \frac{12EI_{zz}}{GA(2h)^2I_y^2} \int_A \left(\frac{\tau_y}{I_z} \right)^2 dA \end{cases} \quad (3)$$

where $\tau_{y/z}$ is the shear force applied to the cross-sectional area. $I_{y/z}$ is the cross-sectional dimension.

Similarly, the constraint vector F^T_3 from the 3-th slice on the 2-th slice can be determined.

In addition to the two constraint vectors, the normal load of the raceway on the slice can be expressed as:

$$q_i = K_{Hertz} \cdot \left\{ (y_i - D_w/2 + C_s^i \quad 0)_{\max} \right\}^{10/9} \quad (4)$$

Where K_{Hertz} denotes Hertzian contact deformation coefficient of the i -th slice, defined as [24]:

$$K_{Hertz} = \pi \cdot (L/n)^{8/9} E' / 3.891^{10/9} \quad (5)$$

where E' is the material-related coefficient, $E' = 1/[(1-\nu_1^2)/E_1 + (1-\nu_2^2)/E_2]$. ν is the Poisson's ratio. E is the Young's modulus. L is the length of the roller.

D_w is the diameter of the roller. C_s^i is the trimming factor for i -th slice, and for logarithmic trimming, it can be expressed as:

$$C_s^i = A \cdot \ln \left[1 - (2x_i/L)^2 \right]^{-1} \quad (6)$$

where A is the factor of logarithmic trimming.

By observing Fig. 2(b), it is clear that the equilibrium equation of the i -th slice can be derived as:

$$F_{i-1}^T + F_{i+1}^T = [0 \quad q_i \quad 0 \quad 0 \quad 0 \quad 0]^T + [0 \quad Q_i \quad 0 \quad 0 \quad 0 \quad 0]^T \quad (7)$$

where F_{i+1}^T is the constraint vector from the $(i+1)$ -th slice on the i -th slice. Q_i is external load assigned to the i -th slice.

By superimposing all external loads on the roller, its equilibrium equation is derived.

$$Q_{External} = \sum_{i=1}^n Q_i \quad (8)$$

After solving Eqs. (6)-(7) by Newton-Raphson approach, the load exerted on each slice can be obtained. Fig. 2(c) provides further details on the solution flowchart. With help of Hertzian line contact theory [24], the stress distribution between each slice and raceway in their interface can be further found by the following equation:

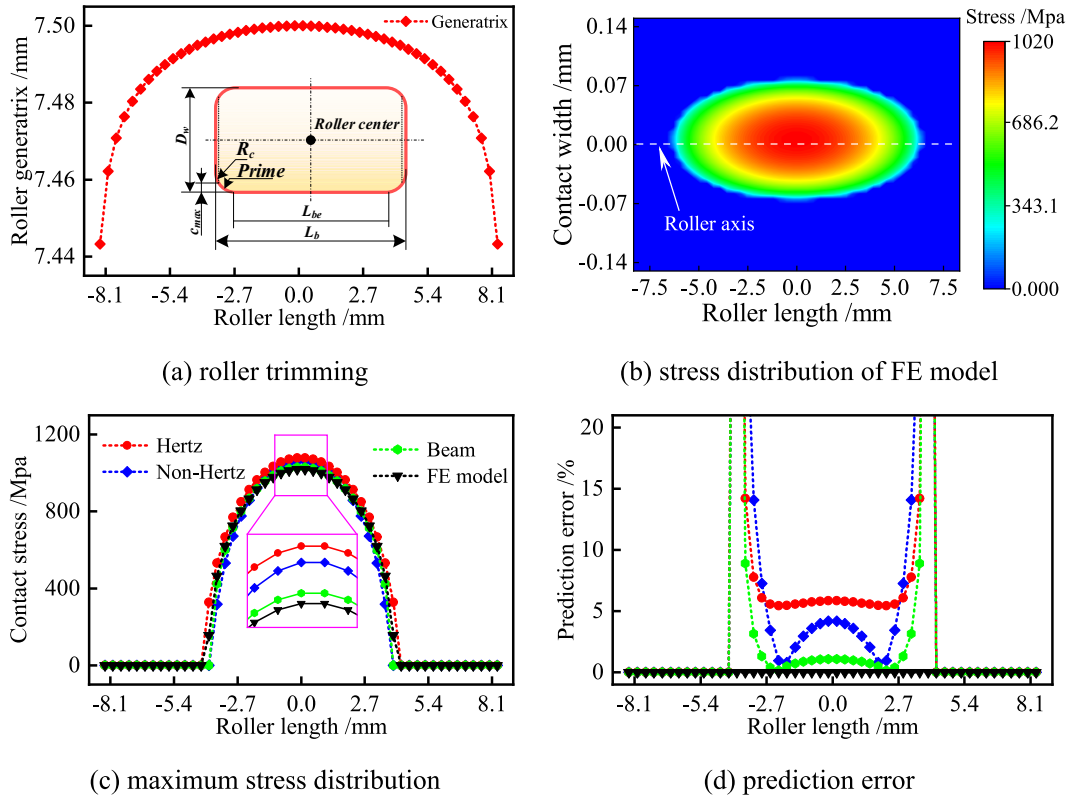
$$p_{ik} = p_{0i} \sqrt{1 - (z_k/a_i)^2} \quad (9)$$

where p_{0i} and a_i are the maximum stress and contact half-width on the i -th slice, respectively, defined as:

Table 2

The detailed parameter of the roller.

Name	Value	Name	Value
Roller diameter/mm	15	Trimming factor	0.0156
Roller length/mm	17	External load/N	1000
Trimming approach	logarithmic	Material	GCr15

**Fig. 3.** Comparison of prediction results for the three slicing methods.

$$\begin{cases} p_{oi} = \sqrt{E'q_i/(\pi R)} \\ a_i = \sqrt{4Rq_i/(\pi E')} \end{cases} \quad (9)$$

where R is equivalent radius of curvature.

2.2. Comparison of different slicing methods

The results of three methods for predicting the stress distribution in the raceway are compared in this section. In addition, the result from the commercial software (Workbench) is also provided to determine the method with highest accuracy.

The analysis object is the roller-raceway contact of cylindrical roller bearing, whose detailed parameters are listed in Table 2. Importantly, only logarithmic trimming was applied to the roller.

Fig. 3 presents the comparison of prediction results for the above three slicing methods. The dimensional parameter of each slice is displayed in Fig. 3(a). The result of stress distribution provided by Workbench is shown in Fig. 3(b). It indicates that only part of the area is stressed, while the rest of the roller and the raceway are not in contact. The results of the stresses in the roller axis in Fig. 3(b) are plotted in Fig. 3(c) along with the predictions from the first three methods.

It can be seen from Fig. 3(c), the results of the four methods are very similar except for the contact zone boundary. Taking the finite element model (FE model) results as a reference, Fig. 3(d) further quantifies the differences among the three methods. Among them, the results of the beam-based slicing method are closest to those of the FE model. It indicates that it has the highest solution accuracy. This is due to the fact that it not only considers the coupling effects between slices, but also treats individual slices as independent rigid bodies. The Hertzian contact-based method, on the other hand, gives the worst results because it ignores the coupling effect between

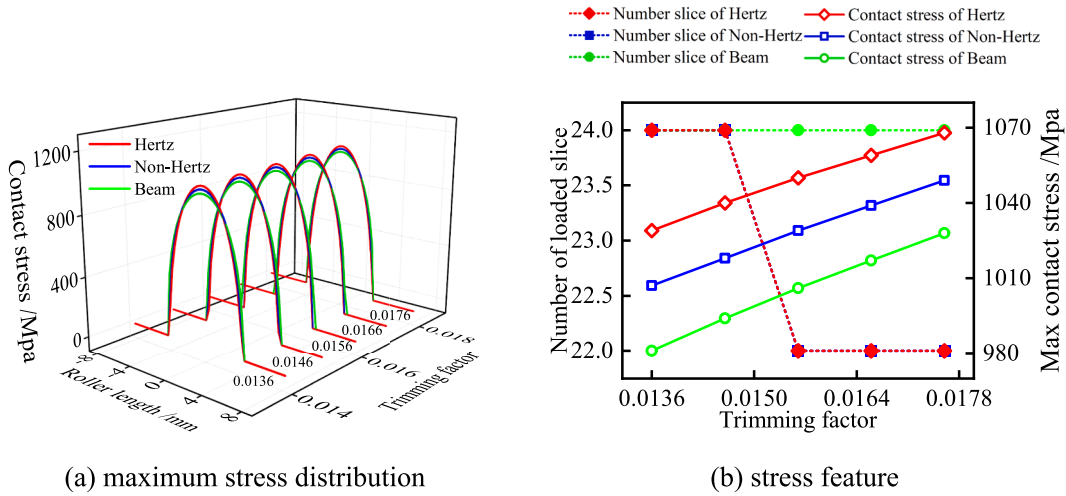


Fig. 4. Effect of trimming factor on the contact feature for three slicing methods.

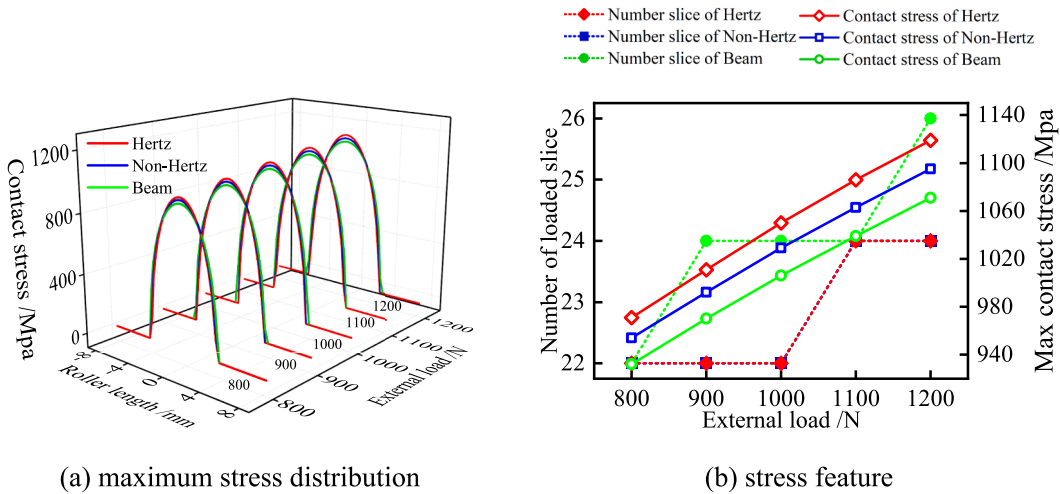


Fig. 5. Effect of external load on the contact feature for three slicing methods.

the slices.

2.3. Studies on roller contact feature

The contact characteristics between roller and raceway are compared for different structural and operating parameters, and the distinctions between the three methods are further emphasized.

(a) Effect of structural parameters.

An external load of 1000 N is applied to the roller and its skew angle is set to 0° . The trimming factor of the roller is gradually increased from 0.0136 to 0.0176, and the contact stress of the roller predicted by the three slicing methods are given in Fig. 4.

By analyzing Fig. 4, it is observed that as the trimming factor increases, the predicted maximum contact stress of the rollers increases for all three slicing methods. The beam-based slicing method predicts the lowest maximum contact stress, while the Hertzian contact-based slicing method predicts the highest. Additionally, with the increase in the trimming factor, the number of loaded slices decreases for both the Hertzian contact-based and influence coefficient-based slicing methods, while it remains constant for the beam-based slicing method. Overall, the beam-based slicing method provides a more uniform stress distribution compared to the other two methods.

(b) Effect of operating conditions.

The effect of operating conditions on the contact features are then investigated. With 0.0156 of trimming factor and 0° of skew angle, the external load is increased from 800 N to 1200 N. The differences in the roller contact features predicted by the three methods are given in Fig. 5.

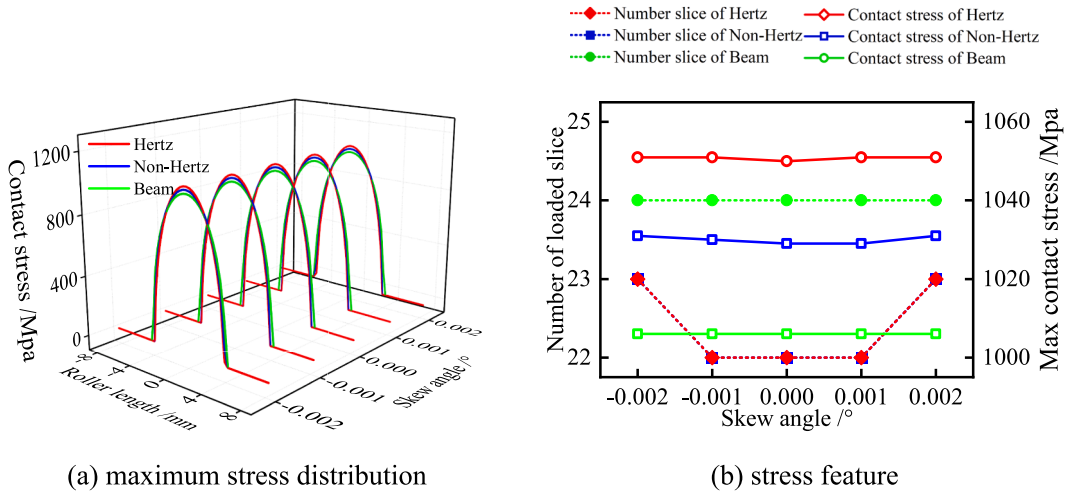


Fig. 6. Effect of roller skew angle on the contact feature for three slicing methods.

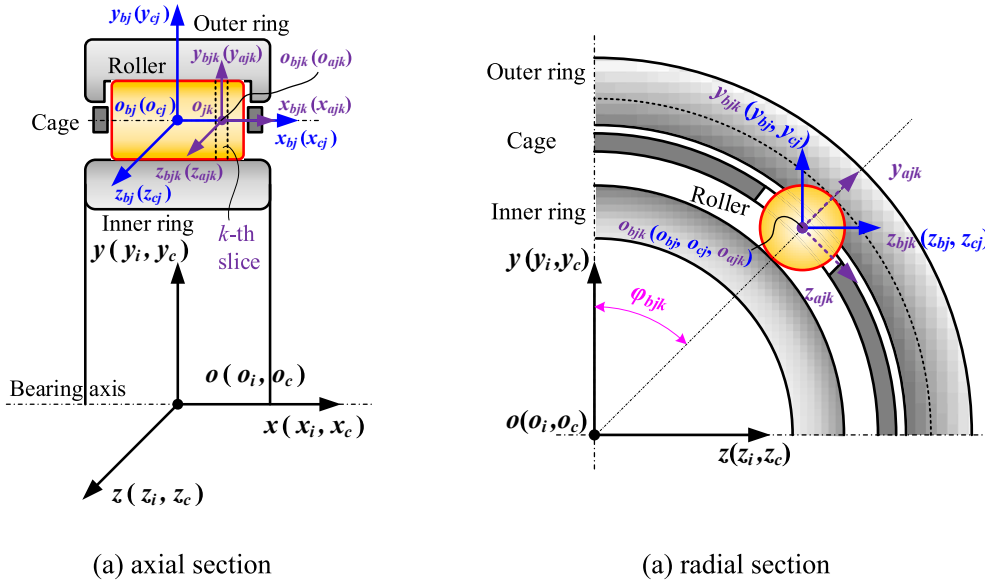


Fig. 7. Schematic of the coordinate frame for bearing dynamic model.

As shown in Fig. 5, both the maximum contact stress and the number of loaded slices grow gradually with the increase of external load. The number of loaded slices increases from 22 to 26 for the beam –based slicing method. The number of loaded slices predicted by the other two slicing methods is similar, increasing from 22 to 24 slices. This indicates that the uniformity of stress distribution predicted by the beam-based slicing method is slightly higher than the other two methods.

Keeping external load (1000 N) and trimming factor (0.0156) constant, the roller skew angle is increased from -0.002° to 0.002° . The results of the three slicing methods for predicting the contact characteristics at different skew angles are further compared, as shown in Fig. 6.

From Fig. 6, it is clear that the skew angle has almost no effect on the maximum contact stress. Among the methods, the maximum contact stress predicted by the beam-based slicing method is consistently the lowest. As the skew angle increases, the number of loaded slices predicted by the slicing methods based on Hertz contact and influence coefficient first decreases, then increases. In contrast, the number of loaded slices predicted by the beam-based slicing method remains constant, likely because it accounts for the minor deformations of the roller.

In summary, regardless of the structural or operating conditions, the beam-based slicing method predicts the highest number of loaded slices and the lowest maximum contact stress, followed by the influence coefficient slicing method.

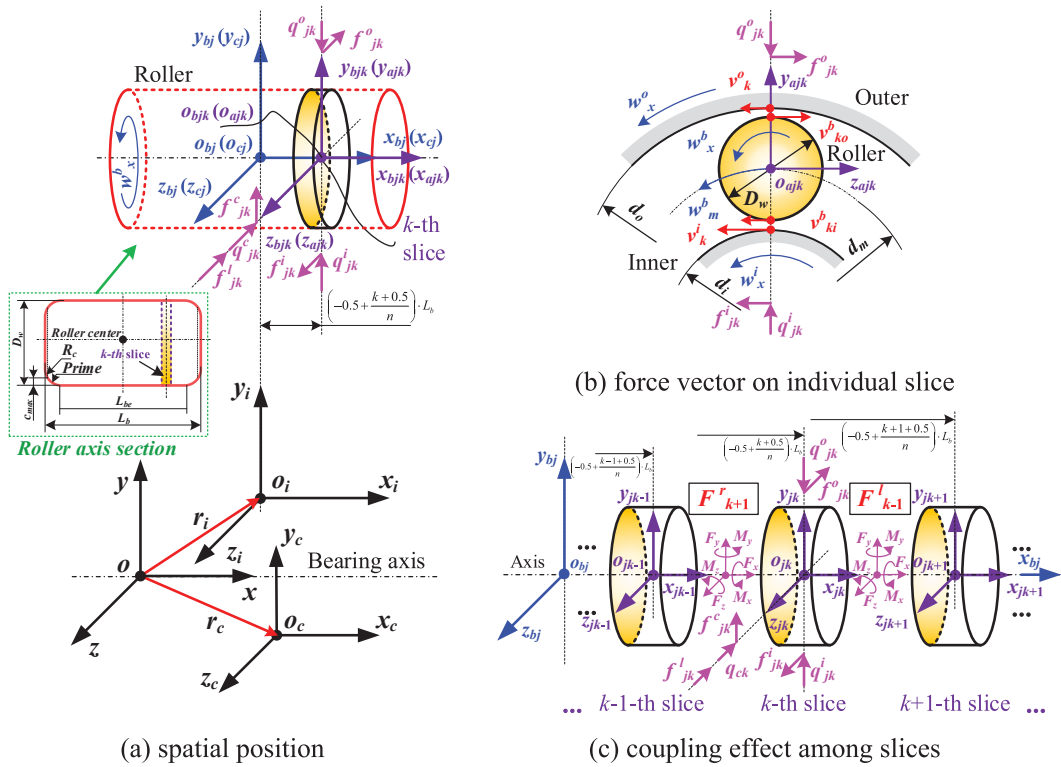


Fig. 8. Schematic of the interaction vector between the roller and raceway.

3. Dynamic modeling of cylindrical roller bearing

In Section 2, the beam-based slicing method is used to analyze the static contact behavior between the roller and the raceway. This method, solved by the Newton-Raphson approach, could not be effectively coupled with existing bearing dynamic models. To address this issue, the method has been improved in this section by treating each slice as an individual rigid body. By concentrating all mass of each slice at its centroid, the same motion differential equations as those for other bearing components are constructed and solved by a differential solver. This enables the flexible roller model to be coupled and solved synchronously with the bearing model while accounting for the roller flexibility. However, it assumes that the entire mass of the slice is concentrated at the centroid, which does not perfectly align with physical reality, leading to potentially questionable results.

Based on the improved slicing method of beam, a new dynamic model of cylindrical roller bearing is presented in this section. Before modeling, a large number of coordinate frames need to be defined to determine the position of each component in space. The interaction vector among the bearing components is further investigated. On this basis, the differential equations for the moving components are constructed and solved. Finally, the dynamic model is validated experimentally.

As shown in Fig. 7, the coordinate frame system consists of seven coordinate frames, defined as: the outer-fixed coordinate frame (o_{xyz}), the inner-fixed coordinate frame ($o_{i-x_i y_i z_i}$), the cage-fixed coordinate frame ($o_{c-x_c y_c z_c}$), the roller-fixed coordinate frame ($o_{bj-x_{bj} y_{bj} z_{bj}}$), the pocket coordinate frame ($o_{cj-x_{cj} y_{cj} z_{cj}}$), the slice coordinate frame ($o_{bjk-x_{bjk} y_{bjk} z_{bjk}}$), the slice azimuthal frame ($o_{ajk-x_{ajk} y_{ajk} z_{ajk}}$). Among them, the outer-coordinate frame is considered as the inertial coordinate frame.

3.1. Force vector exerted on roller

The interaction vector between the roller and raceway is first modelled, displayed in Fig. 8. Due to the uneven loading, the roller needs to be discretized into n slices along its axis. The slicing method of beam is employed to address it. Taking the k -th slice as an example, the force exerted on it is investigated. The spatial position and force excitation of the k -slice are presented in Fig. 8(a). The forces exerted on the slice include internal and external forces, where the internal force is the constraints imposed on it by neighboring slices, and the external forces are those from the raceway, cage, and lubricant. For roller flexibility, its deformation can influence the contact load between the roller and raceway, which in turn affects the oil film thickness and changes the lubrication state. Additionally, the lubricant properties in the contact interface, such as viscosity, may also alter. Fortunately, these effects are implicitly addressed in Wang's traction model [40], which is also adopted in this model.

The focus of this section is to derive the interaction vector for the raceway and lubricant. Firstly, it is assumed that the position vector of the k -slice in the inertial frame is defined as:

$$q_{bjk} = [r_{bjk} \quad \Theta_{bjk}]^T = [x_{bjk} \quad y_{bjk} \quad z_{bjk} \quad \eta_{bjk} \quad \zeta_{bjk} \quad \lambda_{bjk}]^T \quad (10)$$

Its displacement vector with respect to the inner ring in the inner-fixed frame can be further expressed as follows:

$$r_{bjk}^i = [T_i](r_{bjk} - r_i) \quad (11)$$

where r_i is the displacement vector. T_i is the transformation matrix from the inertial to inner-fixed frame, defined by:

$$[T_i] = \begin{bmatrix} \cos\zeta_i \cos\lambda_i & \cos\eta_i \sin\lambda_i + \sin\eta_i \sin\zeta_i \cos\lambda_i & \sin\eta_i \sin\lambda_i - \cos\eta_i \sin\zeta_i \cos\lambda_i \\ -\cos\zeta_i \sin\lambda_i & \cos\eta_i \cos\lambda_i - \sin\eta_i \sin\zeta_i \sin\lambda_i & \sin\eta_i \cos\lambda_i - \cos\eta_i \sin\zeta_i \sin\lambda_i \\ \sin\zeta_i & -\sin\eta_i \cos\zeta_i & \cos\eta_i \cos\zeta_i \end{bmatrix} \quad (12)$$

As shown in Fig. 8(b), the deflection vector between the k -th slice and the inner raceway is then derived as:

$$\delta_{jk}^i = \sqrt{y_{bjk}^2 + z_{bjk}^2} - (d_i - D_w)/2 + C_s^k \quad (13)$$

where d_i is the diameter of the inner raceway.

According to the Hertzian theory, the normal load between the k -th slice and the inner raceway can be calculated by:

$$q_{jk}^i = K_{Hertz} \left\{ (\delta_{jk}^i \quad 0)_{\max} \right\}^{10/9} \quad (14)$$

Due to the presence of lubricating oil, the lubrication state between the roller slice and the raceway is highly complex, involving dry friction, boundary lubrication, mixed lubrication, and elastohydrodynamic lubrication (EHL). As mentioned above, these lubrication states are determined by the combination of film thickness and surface roughness. The calculation of traction forces under the first three lubrication states is relatively simple and has been detailed in Zhang's work [41]. Therefore, the formula for obtaining the traction coefficient between the roller slice and the raceway in the EHL state is presented. Considering computational efficiency, the friction force at the k -th slice/inter raceway interface can be defined based on Wang's work [40].

$$f_{jk}^i = \left[(A + B \cdot |\Delta v_{kj}^i|/u_{kj}^i) e^{C \cdot |\Delta v_{kj}^i|/u_{kj}^i} + D \right] \cdot q_{jk}^i \cdot \text{sign}(\Delta v_{kj}^i) \quad (15)$$

where A , B , C , and D are functions of parameters related to lubrication, contact, material, etc., obtained by:

$$A = A_0 W^{A_1} |W_c/W - 1| U^{A_2} T^{A_3} \quad (16)$$

$$B = B_0 W^{B_1} |W_c/W - 1| U^{B_2} T^{B_3} \quad (17)$$

$$C = C_0 W^{C_1} |W_c/W - 1| U^{C_2} T^{C_3} \quad (18)$$

$$D = D_0 W^{D_1} |W_c/W - 1| U^{D_2} T^{D_3} \quad (19)$$

Where W , U , and T are dimensionless parameters of velocity, load, temperature, etc., respectively. W_c is a function of lubricant temperature. A_x , B_x , C_x , D_x can be determined experimentally. All of the above parameters can be found in Wang's work [40].

Δv_{kj}^i and u_{kj}^i are the relative sliding velocity and rolling velocity between them respectively, defined by:

$$\begin{cases} \Delta v_{kj}^i = v_k^i - v_{ki}^b \\ u_{kj}^i = (v_k^i + v_{ki}^b)/2 \end{cases} \quad (20)$$

where v_k^i is the absolute velocity of the inner ring at the contact area, obtained by:

$$v_k^i = w_x^i \cdot d_i/2 - (w_y^i \cdot \cos\varphi_{bjk}^i + w_z^i \cdot \sin\varphi_{bjk}^i) \cdot x_{bjk}^i \quad (21)$$

where w^i is the angular velocity of the inner ring around the x -axis. φ_{bjk}^i is the azimuth of the k -th slice in the inner-fixed frame.

v_{ki}^b is the absolute velocity of the l -th slice at the contact area, obtained by:

$$v_{ki}^b = w_m^b \cdot \sqrt{y_{bjk}^2 + z_{bjk}^2} - w_{(x)bjk}^i D_w/2 \quad (22)$$

where w_m^b is the orbital velocity of the k -th slice. w_{bjk}^i is the angular velocity of the k -th slice in the inner-fixed frame, $w_{bjk}^i = [T_i] \cdot w_{bjk}$.

Similar to the inner raceway, the interaction vector $[q_{jk}^o, f_{jk}^o]^T$ between the outer raceway and the k -th slice can be obtained.

Due to the presence of lubricant, the churning resistance of the slice in the bearing cavity can be roughly estimated by [22]:

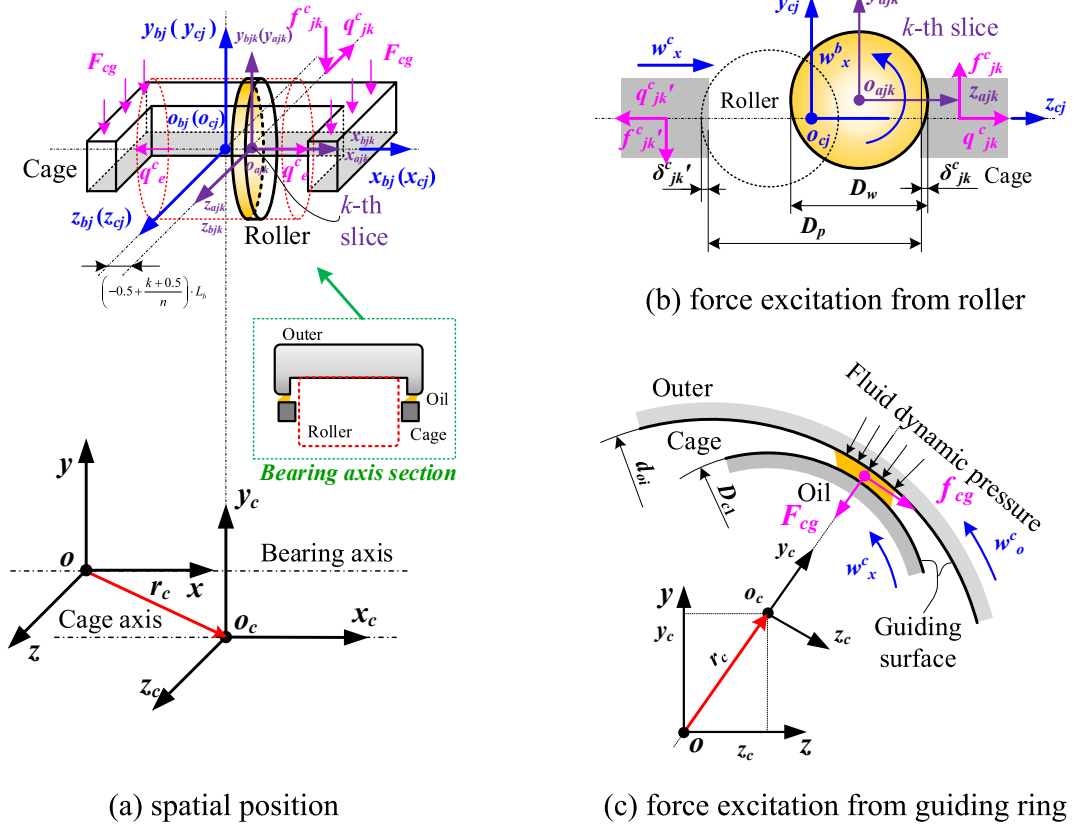


Fig. 9. Schematic of the force excitation vector exerted on the cage.

$$f_{jk}^l = \frac{\pi c_{sv} \rho_v (L/n \cdot D_w \cdot w_{(x)bjk})}{40} \quad (23)$$

where c_{sv} is the drag coefficient. ρ_v is the lubricant density.

As for the load vector $[q_{jk}^c, f_{jk}^c]^T$ from the cage, it will be discussed in detail below.

Based on the model in Section 2.3, the constraint vector of its neighboring slices $([F_{k+1}^l, M_{k+1}^l]^T, [F_{k-1}^l, M_{k-1}^l]^T)$ is further obtained.

Therefore, as shown in Fig. 8(c), the total force vector exerted on the k -th slice in the inertial coordinate frame can be expressed as:

$$F_{jk} = [T_i]^{-1} [T_{ajk}^i]^{-1} \cdot \begin{bmatrix} 0 \\ q_{jk}^i \\ f_{jk}^i \end{bmatrix} + [T_{ajk}]^{-1} \cdot \begin{bmatrix} 0 \\ -q_{jk}^o + f_{jk}^c \\ -f_{jk}^o - q_{jk}^c - f_{jk}^l \end{bmatrix} + [T_{bjk}]^{-1} \left([T_{bj(k-1)}^{bjk}]^{-1} F_{k-1}^l + F_{k+1}^r \right) \quad (24)$$

where T_{ajk}^i is the transformation matrix from the inner-fixed to roller azimuth frame. T_{ajk}^i is the transformation matrix from the inertial coordinate to roller azimuth frame. T_{bjk} is the transformation matrix from the inertial coordinate to roller-fixed frame. $T_{bj(k-1)}^{bjk}$ is the transformation matrix from the k -slice-fixed to $(k-1)$ -th slice-fixed frame. F_{k-1}^l is the constraint vector from neighboring slices.

Similarly, its total moment vector in its fixed body frame can be derived as:

$$M_{jk} = D_w/2 \cdot [T_{bjk}]^{-1} \left([T_i]^{-1} [T_{ajk}^i]^{-1} \cdot \begin{bmatrix} -f_{jk}^i \\ 0 \\ 0 \end{bmatrix} + [T_{ajk}]^{-1} \cdot \begin{bmatrix} -f_{jk}^o - f_{jk}^c \\ 0 \\ 0 \end{bmatrix} \right) + [T_{bj(k-1)}^{bjk}]^{-1} M_{k-1}^l + M_{k+1}^r \quad (25)$$

By observing Fig. 8(a), the vectors of force and moment exerted on the inner raceway from the slices are then derived as:

$$F_i = \sum_{j=1}^Z \sum_{k=1}^n [T_i]^{-1} [T_{ajk}^i]^{-1} \cdot \begin{bmatrix} 0 & -q_{jk}^i & -f_{jk}^i \end{bmatrix}^T \quad (26)$$

$$M_i = \sum_{j=1}^Z \sum_{k=1}^n [T_{ajk}^i]^{-1} \cdot \begin{bmatrix} -d_i/2 \cdot f_{jk}^i & x_{bjk}^i \cdot f_{jk}^i & -x_{bjk}^i \cdot q_{jk}^i \end{bmatrix}^T \quad (27)$$

where Z is the number of the roller.

3.2. Force vector exerted on cage

The role of the cage is to isolate the rolling elements and prevent them from coming into direct contact, which could affect the bearing performance. Therefore, it needs to be considered into the bearing dynamic model as well. The interaction vectors applied on the cage from the other components are displayed in Fig. 9. As shown in Fig. 9(a), these vectors come from the normal load and friction of the rolling element and the guiding force of the ring. The forces from the rolling element are first investigated.

As shown in Fig. 9(b), the deflection between the k -th slice and pocket in the cage rotation direction is defined as:

$$\delta_{jk}^c = \left| z_{bjk}^c \right| - (D_p - D_w)/2 - C_s^k \quad (28)$$

where D_p is the diameter of the cage pocket. r_{bjk}^c is the displacement vector of the k -th slice in the j -th pocket-fixed frame, defined as:

$$r_{bjk}^c = [T_{cj}^c] \cdot [T_c] \cdot (r_{bjk} - r_c) - [0 \quad d_{cp}/2 \quad 0]^T \quad (29)$$

where T_c is the transformation matrix from the inertial to cage-fixed frame. T_{cj}^c is the transformation matrix from the cage-fixed to j -th pocket-fixed frame. d_{cp} is the diameter of the circle where the pocket center is located.

According the theory of Hertzian and coulomb, their normal load and friction can be predicted by the following equations.

$$q_{jk}^c = K_{Hertz} \cdot \left\{ (\delta_{jk}^c \quad 0)_{\max} \right\}^{10/9} \cdot \text{sign}(-z_{bjk}^c) \quad (30)$$

$$f_{jk}^c = \mu_{jk}^c \cdot q_{jk}^c \cdot \text{sign}(-w_x^b) \quad (31)$$

where μ is the frictional coefficient between the pocket and the roller.

In some case, the collision between the end slice and the side beam of the pocket may occur, which can be defined as:

$$q_{ek}^c = K_{Hertz} \cdot \left\{ (x_{bjk}^c - (L_c - L_p)/2 \quad 0)_{\max} \right\}^{10/9} \cdot \text{sign}(x_{bjk}^c) \quad (32)$$

where L_p is the length of the pocket.

Fig. 9(c) presents the fluid dynamic pressure exerted on the cage from the guiding ring, which can be roughly estimated by:

$$F_{cg} = \frac{\eta_0 u_g L_c^3 \epsilon_b^2}{0.25 C_g (1 - \epsilon_b^2)^2} \quad (33)$$

$$f_{cg} = \frac{\pi \eta_0 u_g L_c^3 \epsilon_b^2}{C_g (1 - \epsilon_b^2)^{1.5}} \cdot \text{sign}(v_g) \quad (34)$$

where η_0 is the viscosity of the lubricant. u_g and v_g are the rolling and sliding velocities between the guiding ring and cage. L_c is the cage width. C_g is the guiding clearance. ϵ_b is the relative guiding clearance, defined by:

$$\epsilon_c = \sqrt{y_c^2 + z_c^2} / C_g \quad (35)$$

As shown in Fig. 9(a), the total vectors of the force and moment applied on the cage can be derived as:

$$F_c = [T_c]^{-1} \sum_{j=1}^Z \sum_{k=1}^n [T_{cj}^c]^{-1} \cdot [q_{ek}^c \quad f_{jk}^c \quad q_{jk}^c]^T + [0 \quad -F_{cg} \quad f_{cg}]^T \quad (36)$$

$$M_c = \left\{ \sum_{j=1}^Z \sum_{k=1}^n [T_{cj}^c]^{-1} \cdot \begin{bmatrix} d_{cp}/2 \cdot q_{jk}^c \\ x_{bjk}^c \cdot q_{jk}^c \\ -x_{bjk}^c \cdot f_{jk}^c + d_{cp}/2 \cdot q_{ek}^c \end{bmatrix} + [-d_{c1}/2 \cdot f_{cg} \quad 0 \quad 0]^T \right\} \quad (37)$$

where d_{c1} is the guiding diameter of the cage.

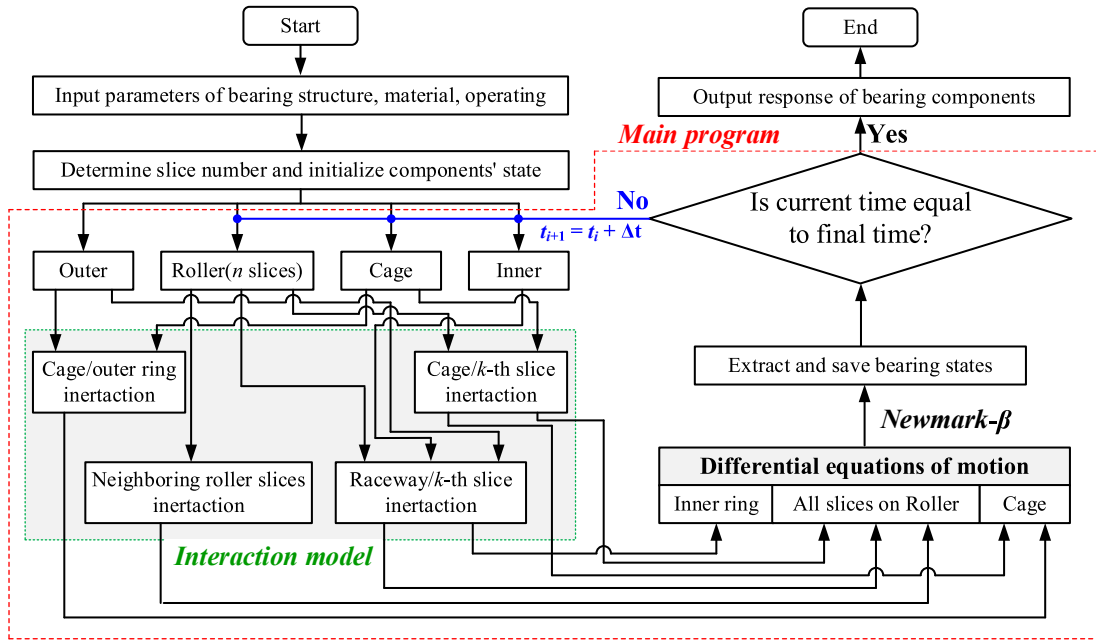


Fig. 10. Flowchart for solving bearing dynamics model based on beam theory.

3.3. Construction and solution of differential equations

Since the internal components of the bearing are considered as rigid bodies, their motion consists of three DOFs of movement and three DOFs of rotation. The movement of the rigid bodies are constructed in the inertial coordinate frame, while the reference coordinate frame of the rotation is the body-fixed coordinate frame. According to Newton-Euler's law of motion, their motion can be determined by these differential equations.

$$m \cdot [\ddot{x} \quad \ddot{y} \quad \ddot{z}]^T = F_{external} \quad (38)$$

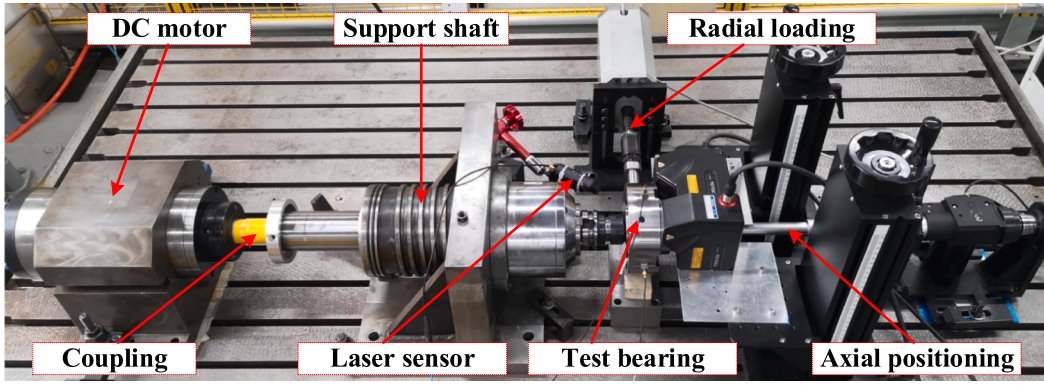
$$\begin{bmatrix} I_x \dot{w}_x - (I_y - I_z) w_y w_z \\ I_y \dot{w}_y - (I_z - I_x) w_z w_x \\ I_z \dot{w}_z - (I_x - I_y) w_x w_y \end{bmatrix} = M_{external} \quad (39)$$

where m is the mass of the component. $[I_x, I_y, I_z]^T$ is the moment vector of inertia. $[\ddot{x}, \ddot{y}, \ddot{z}]^T$ is the linear acceleration vector of the bearing component. $[\dot{w}_x, \dot{w}_y, \dot{w}_z]^T$ is the angular acceleration vector of the bearing component. $[w_x, w_y, w_z]^T$ is the angular velocity vector of the bearing component. $F_{external}$ and $M_{external}$ are the external load vectors applied on the bearing component.

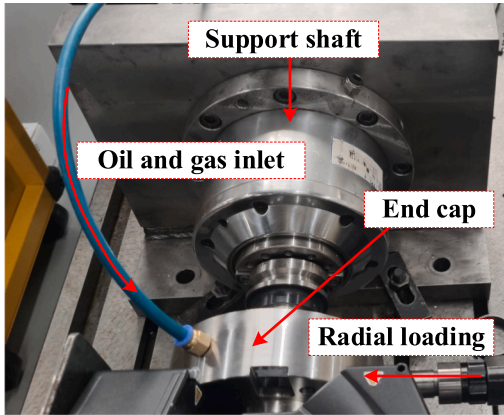
In consideration of the increased complexity discussed above, the calculation time will inevitably rise. The main contributor to the increased computation time is the greater number of differential equations introduced by the roller slice model. Specifically, the number of differential equations grows from $6 \cdot (Z + 1) + 3$ in previous models to $6 \cdot (n \cdot Z + 1) + 3$ in the current model, where n represents the number of slices per roller. The interactions between these slices, combined with the additional complexity from adjacent slice vectors, also significantly add to the computational load. The flowchart for solving the model is further detailed in Fig. 10. At first, the parameters of bearing structure, material and operating needs to be entered into the model. The number of the roller slice is determined and the motion state of the bearing components is initialized. The interaction between the components in Section 3 is used to convert this state into the force vectors, which is the basis for differential equations of each component. These equations are solved by the Newmark- β algorithm to obtain the transient motion of the bearing components at the moment. It is then determined if the set time has been reached. If not, the state at that moment is used as the initial value for the next iteration, otherwise the loop is skipped and the bearing simulation results are entered.

3.4. Model validation

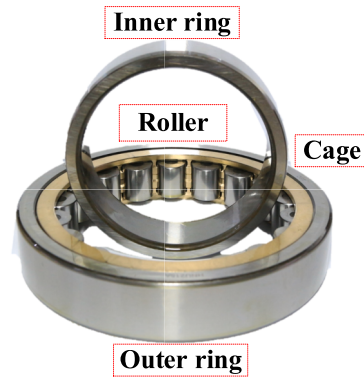
A test bench for bearing dynamic performance was built to validate the dynamic model in Section 3. The main structure of the bench and auxiliary systems are described in detail in Fig. 11. As shown in Fig. 11(a), the test bench consists of several parts such as driving motor, supporting shaft system, testing system, loading system and so on. Since the test bearing is lubricated with oil, it is necessary to provide it with lubricant to ensure its operational stability. Fig. 11(b) further depicts the structure of the loading and



(a) structure diagram for whole test rig



(b) loading and lubrication



(c) test bearing

Fig. 11. Schematic diagram of dynamic performance test bench for cylindrical roller bearings.

Table 3

List of main structural parameters for NU215.

Name	Value
Diameter of inner raceway d_i /mm	88.5
Diameter of outer raceway d_o /mm	118.57
Bearing width B /mm	25
Roller diameter D_w /mm	15
Roller width L /mm	17
Number of roller Z	18
Pocket diameter D_p /mm	16.4
Pocket Width L_p /mm	17.3
Cage width L_c /mm	24.2

lubrication. The cage speed is obtained by using a laser sensor to monitor the passing frequency of the reflective strip on the cage. However, due to limitations in the time scale of the measurement, critical frequencies in the cage speed spectrum may be obscured. Therefore, only the experimental and predicted cage speeds are compared to validated the bearing dynamic model. The external structure of the test bearing is shown in Fig. 11(c) and consists of inner ring, outer ring, cage and rollers.

The test bearing was identified as NU215 and its detailed structural parameters are listed in Table 3. Driving speeds of 2400 r/min and 3600 r/min are applied to the test bearings and its radial load is gradually increased from 500 N to 3000 N. The cage speeds provided by the test and model are presented in Fig. 12.

As shown in Fig. 12, regardless of the driving speed, significant sliding occurs under smaller radial loads. With the gradual increase of radial load, the cage speed increases, which indicates a suppression of sliding inside the bearing. This is because the cage speed represents the average orbital speed of all rolling elements in the bearing. A higher cage speed means that the orbital speed of the rolling elements has increased, reducing the relative sliding velocity between the rolling elements and the raceway, thereby mitigating

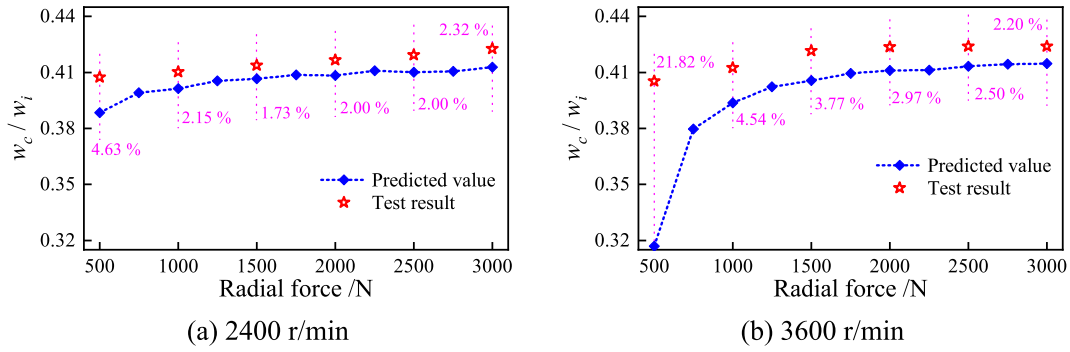


Fig. 12. Comparison between the test results and the model predictions.

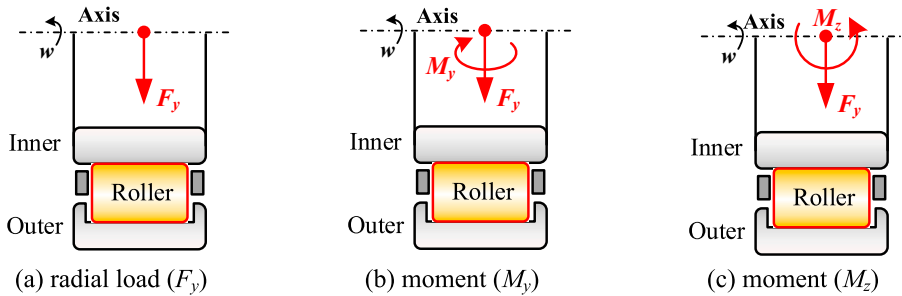


Fig. 13. Different loading method for cylindrical roller bearings.

sliding within the bearing. Under the same radial load, sliding inside the bearing is much more severe at high speeds than at low speeds.

Most importantly, the deviation between the model-predicted cage speeds and the experimentally measured cage speeds does not exceed 5 %, except for a few outliers. This strong agreement effectively validates the bearing dynamic model presented in Section 3. The observed deviations may be attributed to measurement errors, particularly in cases where the radial load is very small, as the loading device may struggle to accurately apply such small loads.

Unfortunately, the current experimental setup does not allow for a detailed comparison of time and frequency domain responses. Specifically, the laser speed sensor used to measure the cage speed has a larger sampling scale than the inherent frequency components of the cage motion, which limits the ability to perform frequency domain analysis. Furthermore, due to the floating structure of the bearing housing, it was not possible to capture the vibration frequency of the bearing rings. Nevertheless, the comparison of average cage speeds under stable conditions still provides a reliable validation of the model, consistent with common practices in bearing dynamic modeling [11,17,22].

4. Investigation of bearing dynamic behavior

In this section, the dynamic performance of cylindrical roller bearings is investigated under three typical operating conditions. The specific loading methods for the bearing are presented in Fig. 13. The indexes of bearing performance include: contact load, contact stress, sliding speed and PV factor. The relationship between the sliding velocity and the angular speed of the roller is explored.

4.1. Effect of y-axis load

Firstly, the effect of radial load on the bearing dynamic performance was investigated. By setting the drive speed to 5000 r/min and increasing the radial load exerted on the bearing from 500 N to 5000 N, some performance indexes between the roller and raceway can be obtained, including contact load, contact stress and sliding velocity.

(a) contact feature.

Fig. 14 presents the contact feature of the inner raceway under different radial loads (y-axis). As shown in Fig. 14(a), with the increasing radial load, the magnitude and frequency of change in contact load between the inner raceway and roller increases. These contact loads in frequency domain are displayed in Fig. 14(b) to further investigate the effect of radial load on them. By observing Fig. 14(b), the richness of the frequency components in the contact load variation decreases with increasing radial load. It is eventually dominated by the frequency components of the first few orders. This is due to the fact that lighter radial loads can lead to severe sliding within the bearing, which results in its extremely unstable performances. As the radial load increases, the sliding inside the bearing is

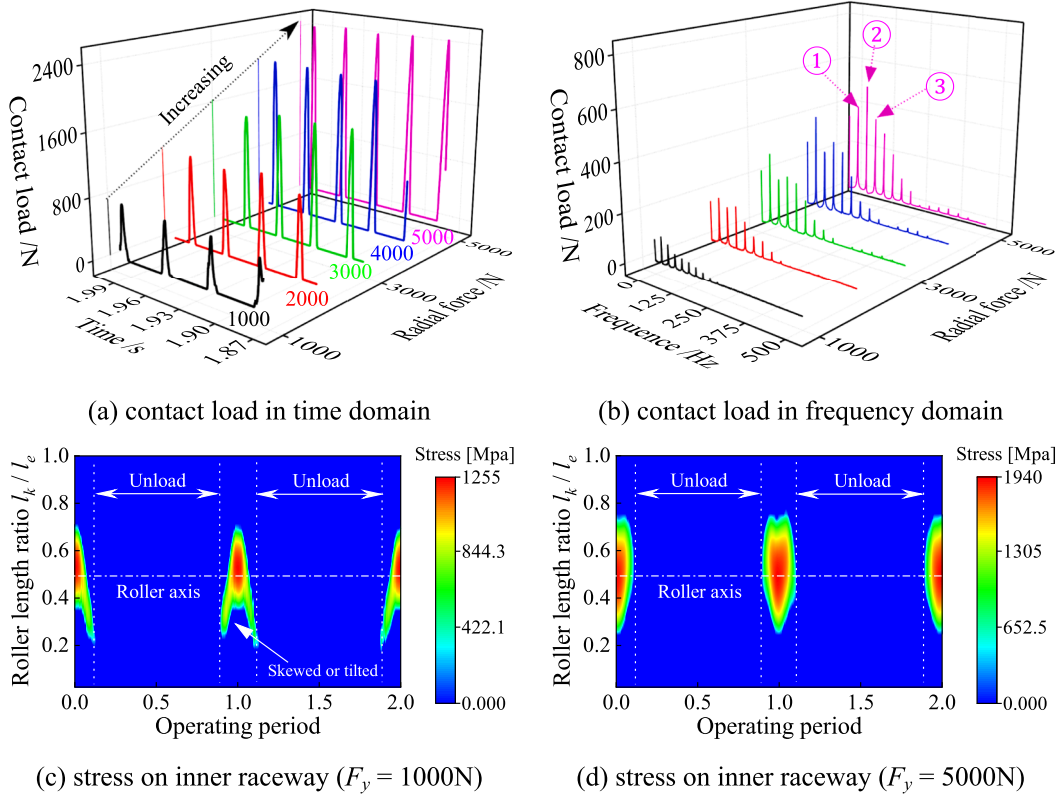


Fig. 14. Effect of y-axis load on contact feature of inner raceway.

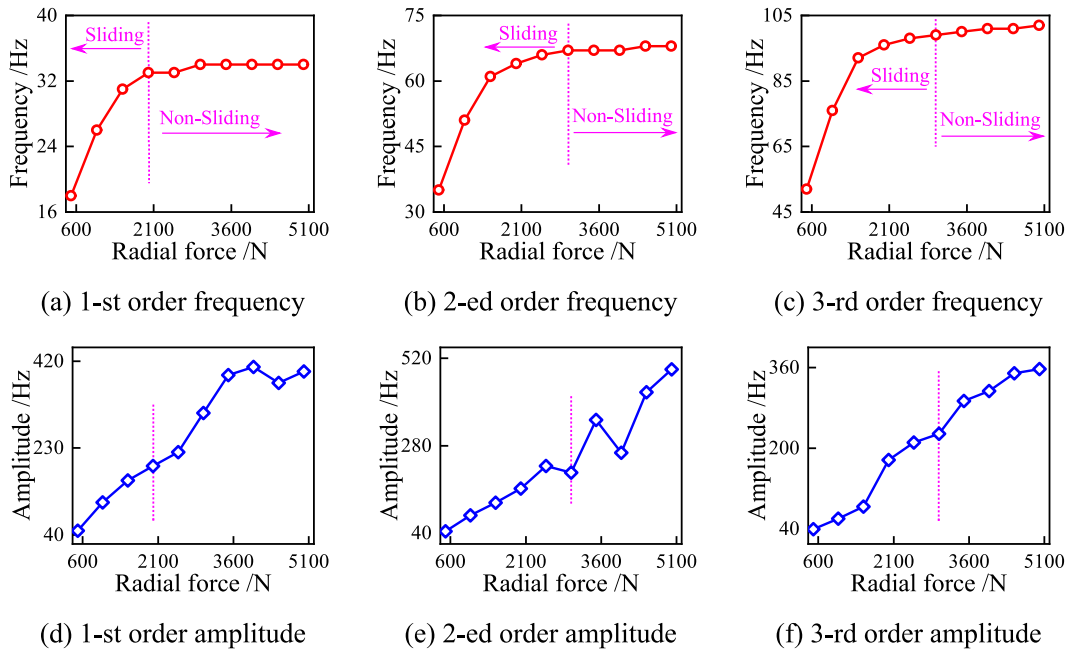


Fig. 15. Variation of frequency and amplitude for contact load on inner raceway (F_y).

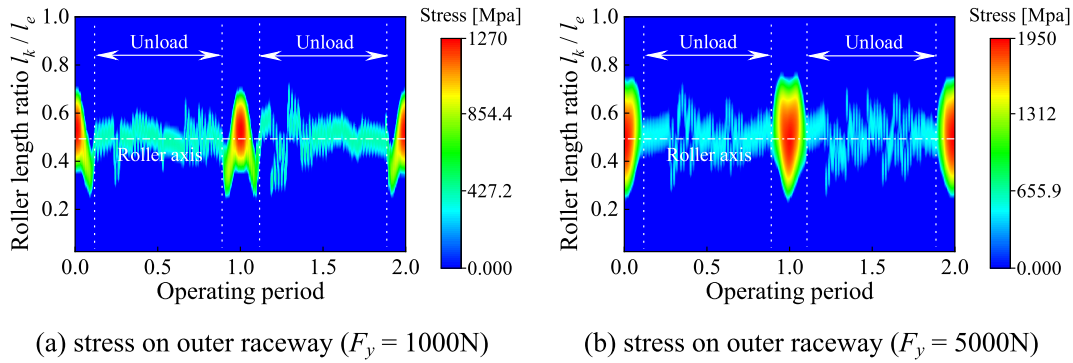


Fig. 16. Effect of y-axis load on contact feature of outer raceway.

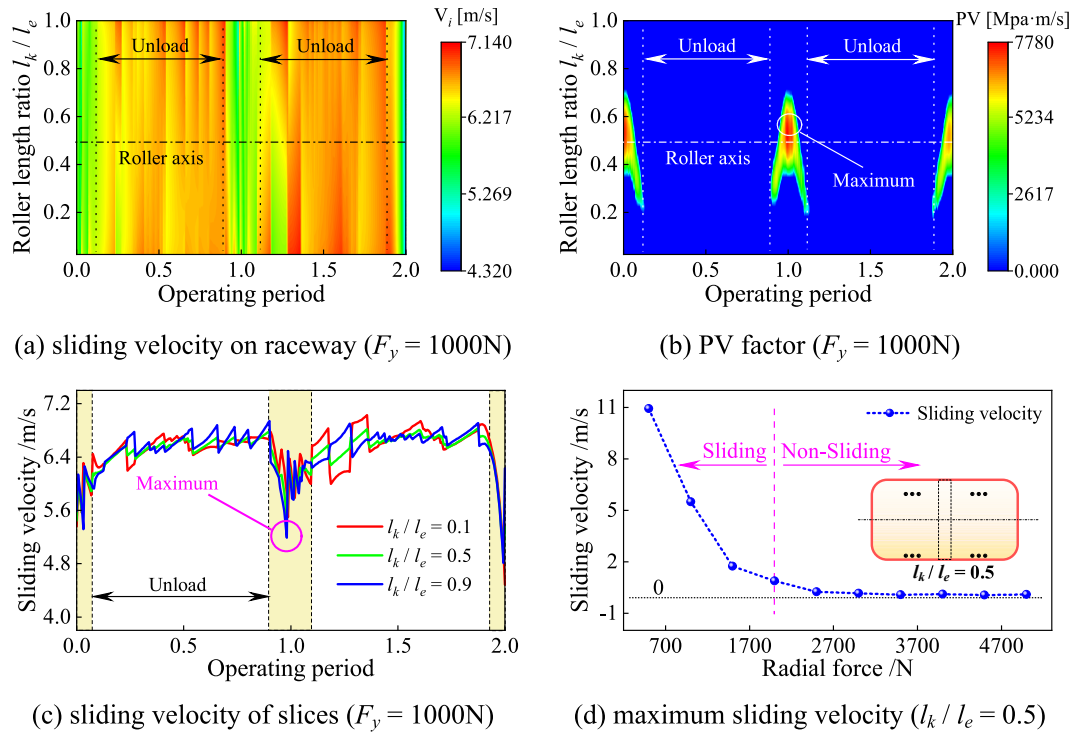


Fig. 17. Effect of y-axis load on sliding feature of inner raceway.

alleviated and its stability strengthens. The contact stress in the inner raceway is further given in Fig. 14(c) and (d). Since the roller is logarithmically trimmed, there is no stress between the roller edge and the inner raceway. As shown in Fig. 14(c), the light radial loads not only cause severe bearing sliding, but also tilt and deflect the rollers. The point of maximum contact stress thus deviates from the roller center. From Fig. 14(d), it can be seen that with the increase of radial load, the tilt and skew of the roller decreases gradually. As a result, the point of maximum contact stress between the roller and inner raceway gradually approaches its center.

Fig. 15 shows the frequency and amplitude of the first three orders of contact loads for different radial loads. As shown in Fig. 15(a), (b) and (c), the frequency of these loads is an integer multiple of the frequency of roller passage inside the bearing. Sliding is also reflected in the frequency of contact load variations. The higher the order, the heavier the radial load required to eliminate the bearing sliding. Therefore, it can be assumed that the instability of the load fluctuation is excited by the passing frequency of the rolling element. Unsimilar to the frequency, as shown in Fig. 15(d), (e) and (f), their amplitudes have been trending upward.

In addition, the contact stress between the roller and outer raceway is depicted in Fig. 16. The contact stress on the outer raceway changes in the same way as in the inner raceway, except in the non-loaded regions. Due to the load excitation from the outer and cage in the non-loaded region, the rolling element has highly dynamic characteristics. As a result, the stresses in this region take on a high-frequency sawtooth shape, which is due to tilting and deflecting motion of the roller.

(b) sliding feature.

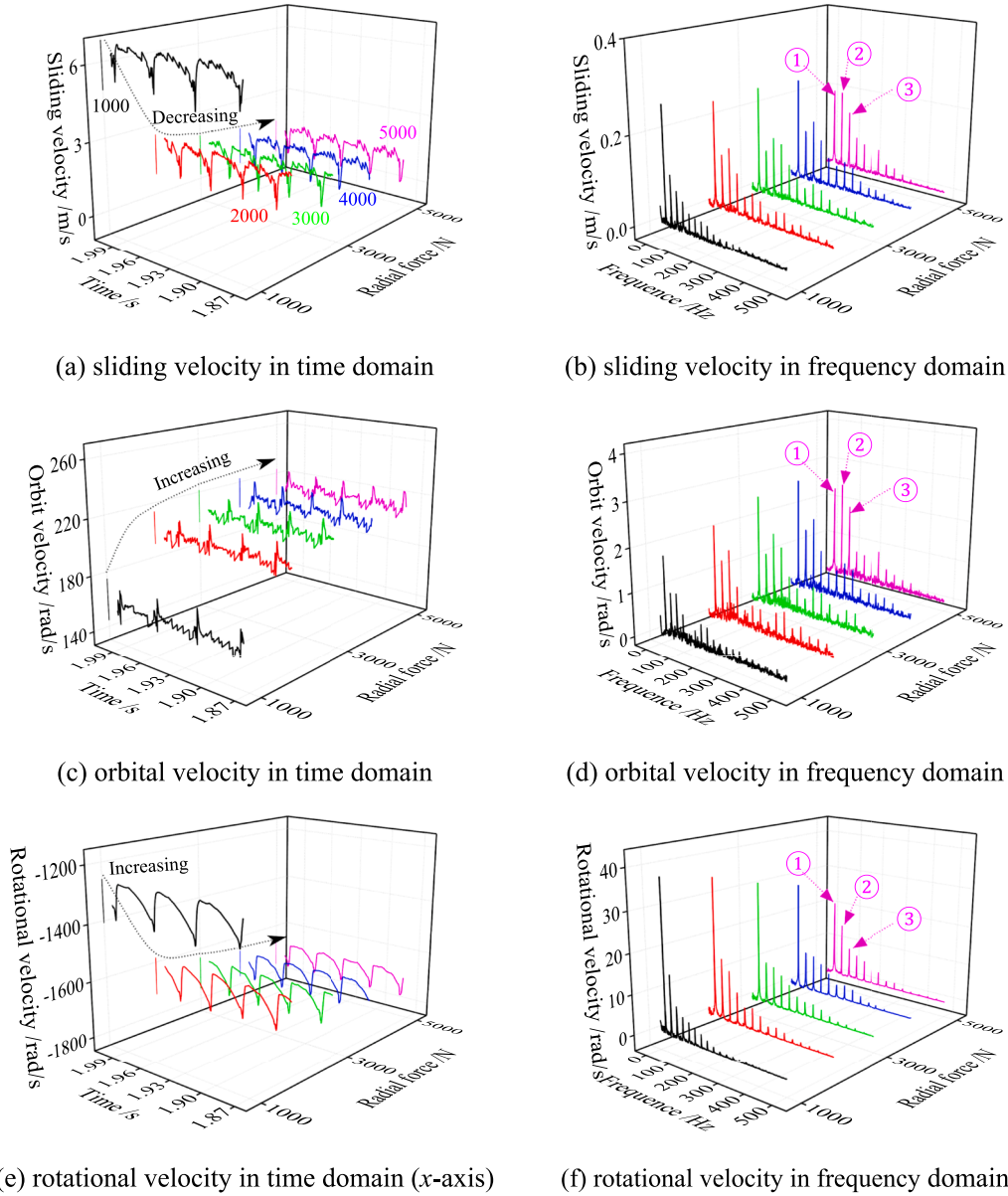


Fig. 18. Relationship between sliding velocity and rotation speed (F_y).

The sliding feature on the inner raceway with different radial loads is displayed in Fig. 17. As shown in Fig. 17(a), the sliding velocity of the inner raceway is up to 7 m/s under a radial load of 1000 N. In the loaded region, the sliding velocity decreases and then increases. When the roller is located in the unload region, the random variation of sliding velocity is reflected not only in the azimuthal direction, but also in its axis direction. Fig. 17(b) presents the PV factor between the roller and inner raceway under that radial load, which is an important performance index. The maximum PV factor is further away from the roller axis than the contact stress. It indicates that the roller is heavily deflected in the load zone. The curves of sliding velocities at the ends and center of the roller are further shown in Fig. 17(c) to investigate the effect of roller tilt on sliding velocity. In the unloaded region, the sliding velocity is affected by the roller tilt and fluctuates in a jagged manner; in the loaded region, the sliding velocity dispersion between the slices along the roller axis is not obvious. As shown in Fig. 17(b) and (c), the maximum PV occurs near the maximum contact stress where the sliding velocity is minimized. The effect of radial load on the sliding velocity of the middle slice ($l_k / l_e = 0.5$) is displayed in Fig. 17(d). As the radial load increases, the sliding velocity on the raceway decreases, which proves that the heavy radial load helps to inhibit bearing sliding. However, it should be noted that excessive radial loads can shorten the bearing fatigue life.

Once the driving speed of the bearing is fixed, the sliding velocity between the roller and raceway is determined by the angular velocity and the orbital velocity of the roller together. Therefore, the relationship between them is presented in Fig. 18 to investigate

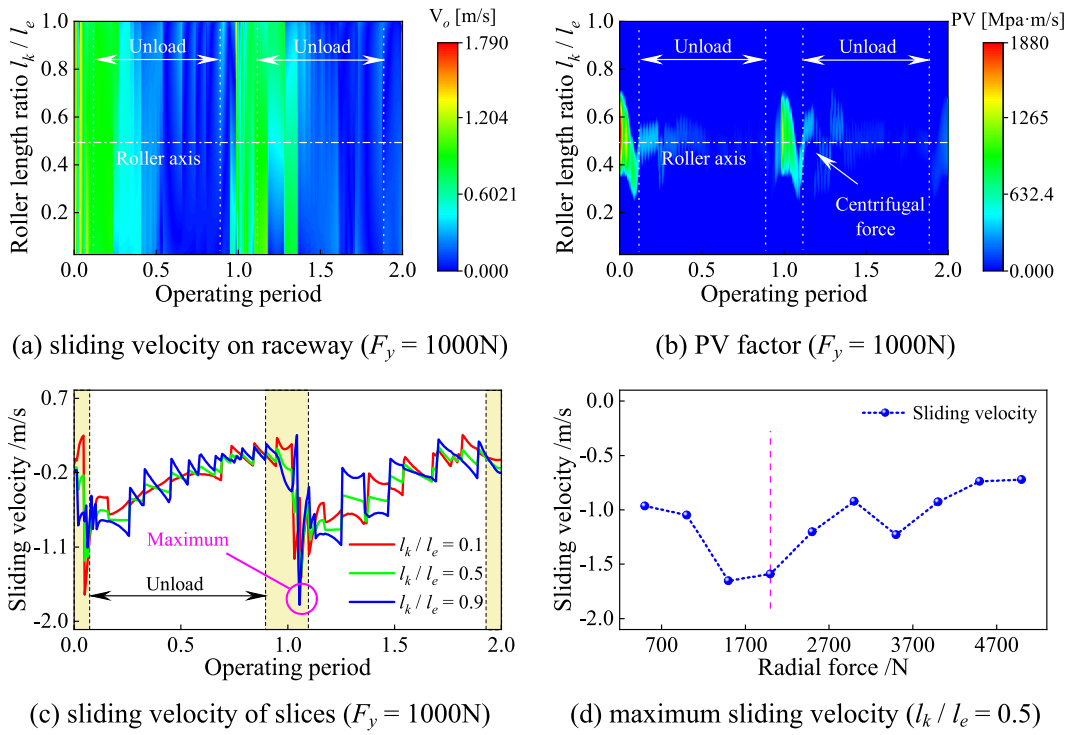


Fig. 19. Effect of y-axis load on sliding feature of outer raceway.

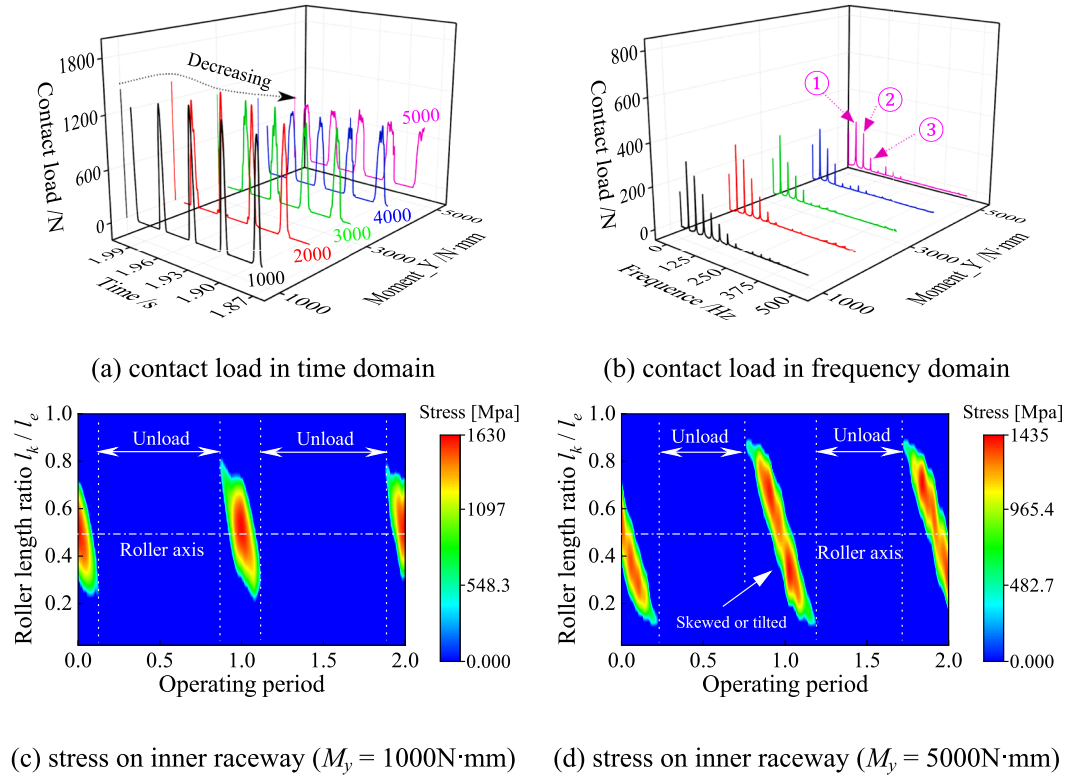


Fig. 20. Effect of y-axis moment on the contact feature of the inner raceway.

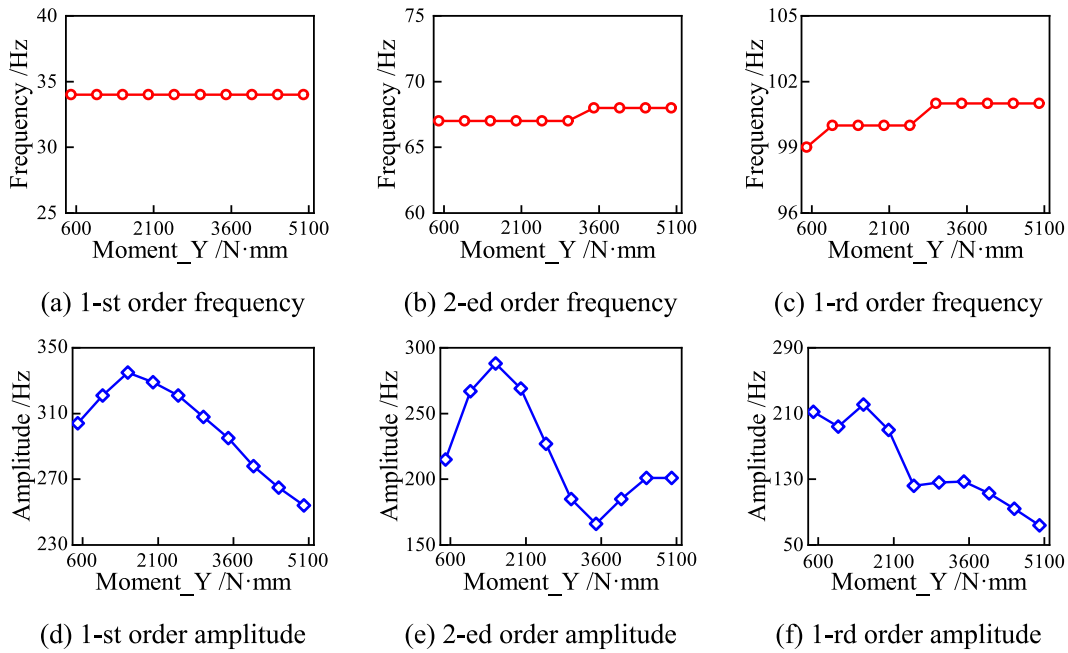


Fig. 21. Variation of frequency and amplitude for contact load on inner raceway (M_y).

the effect of radial load on the time-varying features of the sliding velocity. As shown in Fig. 18(a) and (b), when the radial load is light, the sliding velocity inside the bearing is dominated by the first-order roller passage frequency in the frequency domain. As the radial load increases, the amplitude of the first three order frequencies decreases significantly and the richness of the main frequency spectrum increases. For the angular velocity of the roller in Fig. 18(c) and (d), as the radial load increases, the amplitude of the first three orders of frequencies increases and the richness of their spectra increases. In Fig. 18(e) and (d), the trend of orbital velocity is the same as the spectrogram of sliding velocity. It suggests that the sliding velocity between the roller and inner raceway is strongly influenced by the orbital velocity of the roller. Therefore, for bearings that frequently operate under sliding conditions, maximizing roller orbital velocity will help mitigate this damage. In addition, once the sliding inside the bearing is inhibited, the fluctuation of the sliding velocity depends mainly on the angular velocity of the roller.

Additionally, the sliding between the roller and outer raceway should not be overlooked, although it is relatively small compared to the inner raceway. The effect of radial load on the sliding velocity of the outer raceway is shown in Fig. 19. As shown in Fig. 19(a) and (b), the high probability of wear on the outer raceway occurs in the second half of the load zone due to the high PV factor and sliding velocity in this region. It is worth noting, however, that wear of the outer raceway in the non-loaded region due to centrifugal force should not be overlooked, as it may slide at high speeds, resulting in elevated PV factors. It can be seen from Fig. 19(c), the fluctuations of the sliding velocity in the roller axis direction are more significantly affected by roller tilt in the outer raceway than in the inner raceway. In addition, the azimuthal angle at which the maximum sliding velocity occurs in the outer raceway lags behind that of the inner raceway. From Fig. 19(c), the radial load has little effect on the sliding velocity of the outer raceway, which fluctuates randomly with the radial load.

4.2. Effect of y-axis moment

Secondly, the effect of y-axis moment on the bearing performance was studied. The driving speed of the bearing is kept at 5000 r/min, and the radial load of 3000 N is exerted on it. As shown in Fig. 13(b), the moment on y-axis is increased from 500 N·mm to 5000 N·mm, the dynamic performance between the roller and raceway can be further obtained.

(a) contact feature.

The contact load between the roller and inner raceway under different moments (y-axis) is presented in Fig. 20. As shown in Fig. 20(a), with the gradual increase of the y-axis moment, the peak contact load between the roller and the inner raceway decreases. However, the cyclic curve of the load becomes less smooth. Fig. 20(b) further displays the contact load of the inner raceway in frequency domain. As the y-axis moment becomes heavier, the richness of the spectrogram fades away, creating a variation dominated by first-order frequency components. The stress distribution of the roller in contact with the inner raceway for two typical operating conditions is given in Fig. 20(c) and (d), respectively. The increase in the y-axis moment helps to expand the contact area and reduce the maximum contact stress in the inner raceway. Interestingly, the maximum contact stress in the load region increases from one to two during this process, which are symmetrically distributed around the roller center.

Fig. 21 shows the variation of frequency and amplitude for contact load on inner raceway. As shown in Fig. 21(a), (b) and (c), the

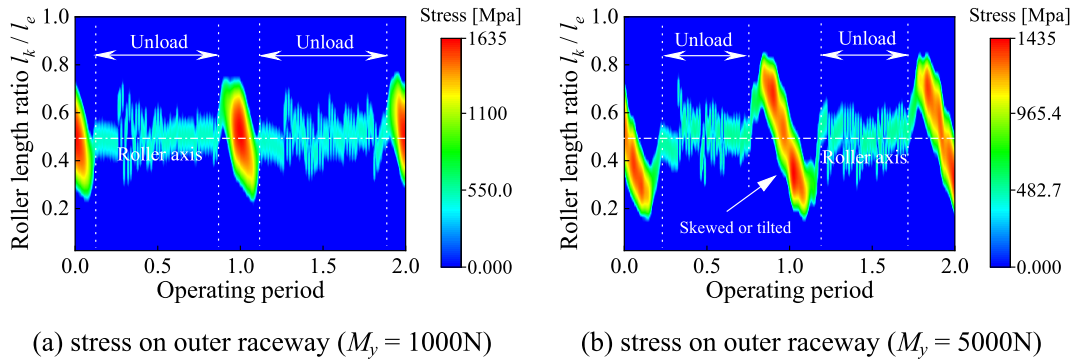


Fig. 22. Effect of y-axis moment on contact feature of outer raceway.

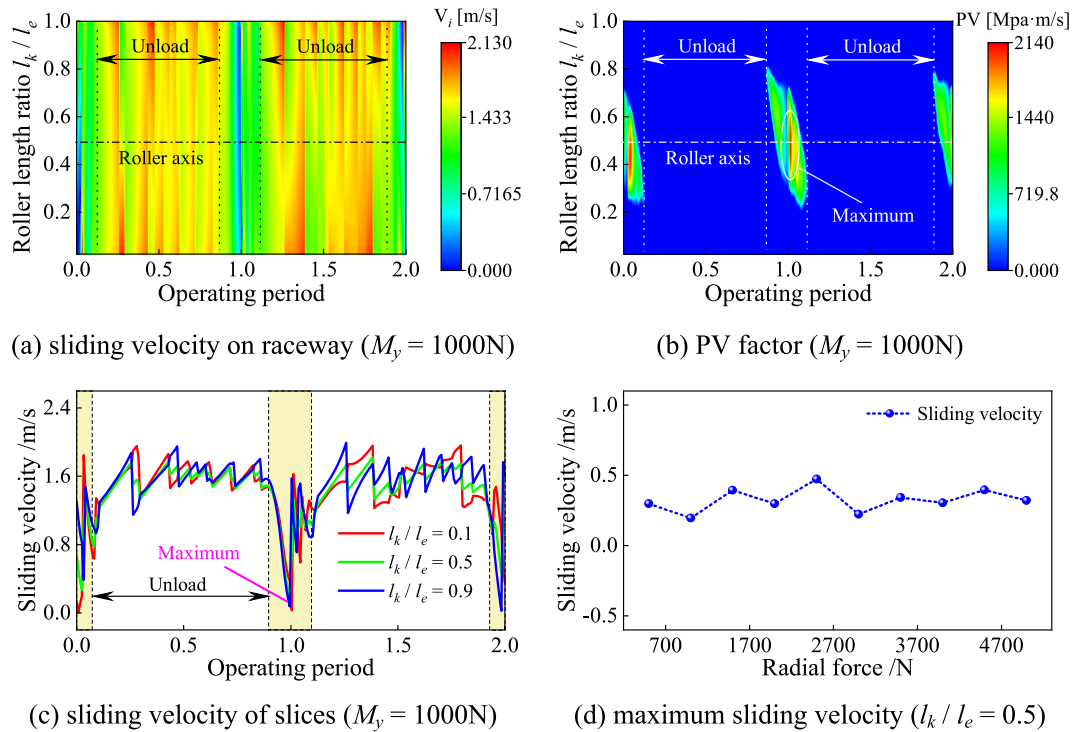


Fig. 23. Effect of y-axis moment on sliding feature of inner raceway.

radial load has less effect on the first and second order frequencies but more effect on the third order frequency. From Fig. 21(d), (e) and (f), it can be seen that three distinct trends were shown for the first three orders of amplitude. With the increase of the y-axis moment, the amplitude of the first-order increases and decreases. As for the third-order, it has been in a downward process.

The variations of the contact stress on the outer raceway are further given in Fig. 22. Larger loaded region on the outer raceway can be observed compared to the inner raceway. In addition, the maximum contact stress in the outer raceway is greater than in the inner raceway due to centrifugal forces. The contact stress between the roller and outer raceway shows sawtooth-like random fluctuations in the non-loaded region due to the roller deflection. In the loaded region, on the other hand, the deflection of the roller is well suppressed, and its stress with the outer raceway shows a well regularity variation.

(b) sliding feature.

Fig. 23 presents the effect of the y-axis moment on sliding feature of the inner raceway. As shown in Fig. 23(a), the sliding velocity on the inner raceway is numerically floated both circumferentially and axially. The values of sliding velocity in the unloaded region are significantly larger than those in the loaded region. When the contact force reaches its maximum value, the sliding velocity is at its minimum value. Under the combined effect of contact stress and sliding velocity, the maximum value of PV factor occurs after the maximum contact stress, as shown in Fig. 23(b). By Comparing Fig. 17(c) with Fig. 23(c), the y-axis moment significantly increases the difference in sliding velocity between the loaded region and the un-loaded region. Fig. 23(d) indicates that the effect of the y-axis

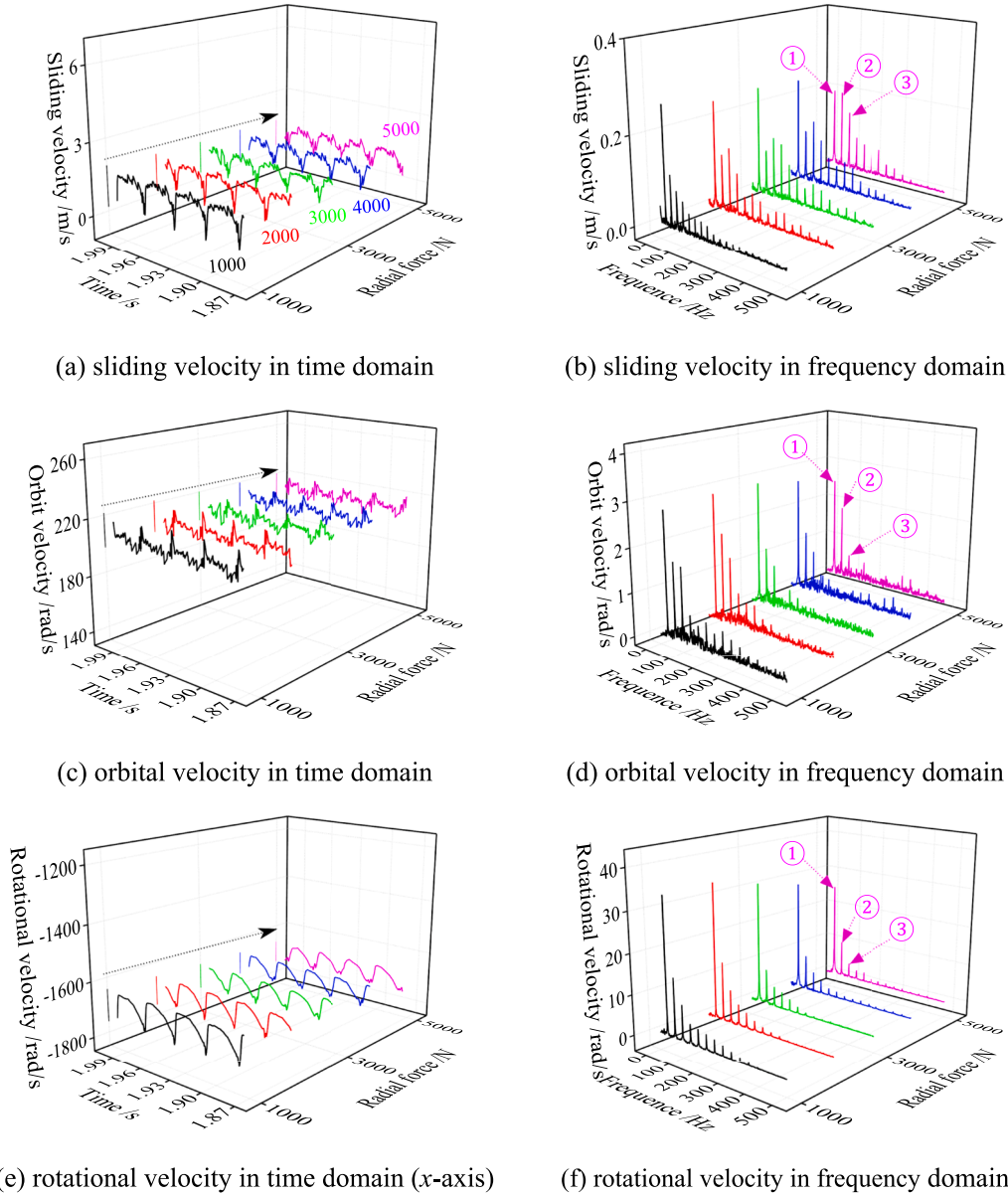


Fig. 24. Relationship between sliding velocity and rotation speed (M_y).

moment on the maximum sliding velocity on the inner raceway is insignificant. This may be due to the fact that there is no serious sliding inside the bearing under a radial load of 3000 N.

In order to further investigate the sawtooth fluctuation feature of the velocity in Fig. 23(c), the variation of roller velocity in time and frequency domains for different y-axis moments is presented in Fig. 24. As shown in Fig. 24(a) and (b), the amplitude of the first-order frequency decreases with the increase of the moment. Heavy moments enrich the spectrogram of sliding velocities on the inner raceway. On the contrary, Fig. 24(c), (d), (e) and (f) indicate that the spectrogram of the roller speeds tends to poorer as the moment increase. Additionally, the amplitude of the first-order frequency shows a same trend as the sliding velocity of the inner raceway.

Fig. 25 further displays the effect of the y-axis moment on the sliding feature of outer raceway. As shown in Fig. 25(a), the sliding velocity starts to decrease near the entrance of the loaded region and then increase at its exit, which is the opposite trend of the inner raceway. An interesting phenomenon can be observed in Fig. 25(b) that only half of the loaded regions have large PV values, with the largest PV value occurring in the area of highest stress. In the same way as the velocity variations on the typical slices of the inner raceway, the sliding velocities of the outer raceway in Fig. 25(c) also showed a sawtooth-like variation. As shown in Fig. 25(d), the y-axis moment has less effect on the sliding velocity in the middle slice of the outer raceway.

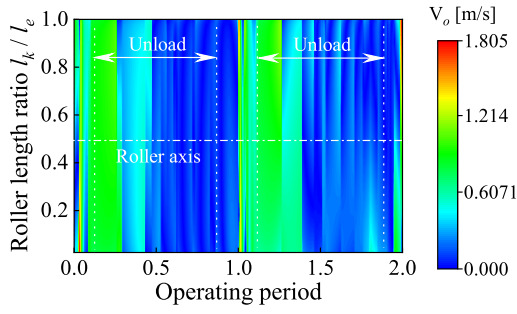
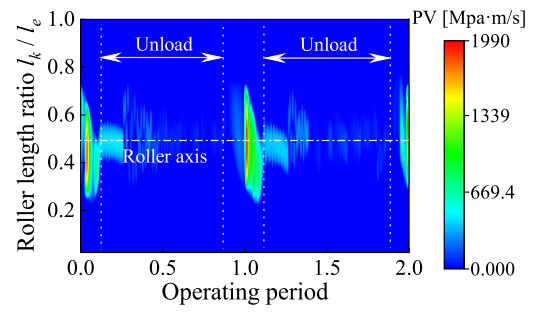
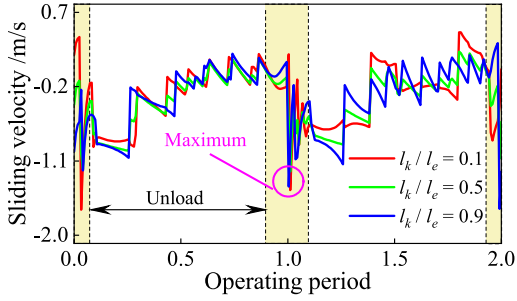
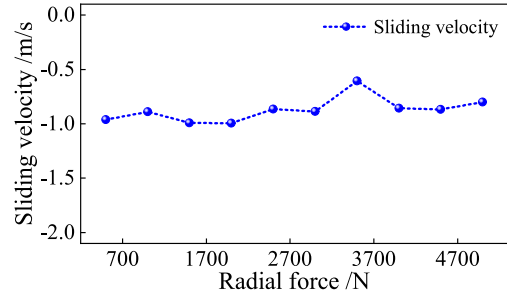
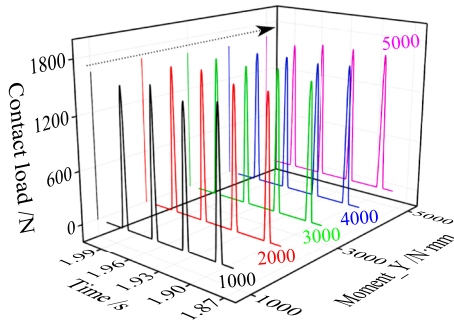
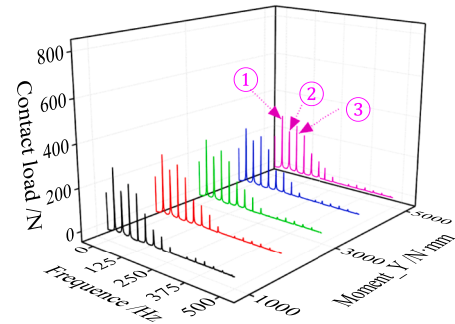
(a) sliding velocity on raceway ($M_y = 1000\text{N}$)(b) PV factor ($M_y = 1000\text{N}$)(c) sliding velocity of slices ($M_y = 1000\text{N}$)(d) maximum sliding velocity ($l_k / l_e = 0.5$)

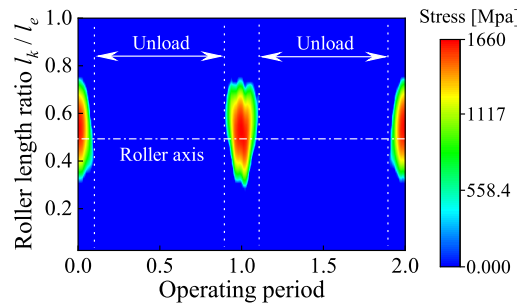
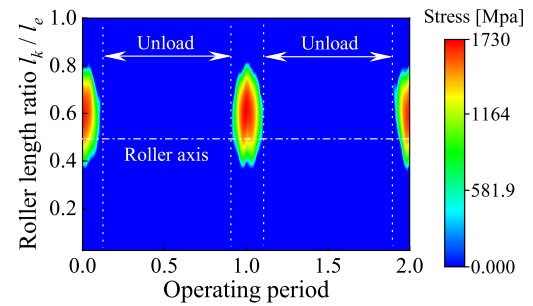
Fig. 25. Effect of y-axis moment on sliding feature of outer raceway.



(a) contact load in time domain



(b) contact load in frequency domain

(c) stress on inner raceway ($M_z = 1000\text{N}\cdot\text{mm}$)(d) stress on inner raceway ($M_z = 5000\text{N}\cdot\text{mm}$)Fig. 26. Effect of moment on the contact feature of the inner raceway (M_z).

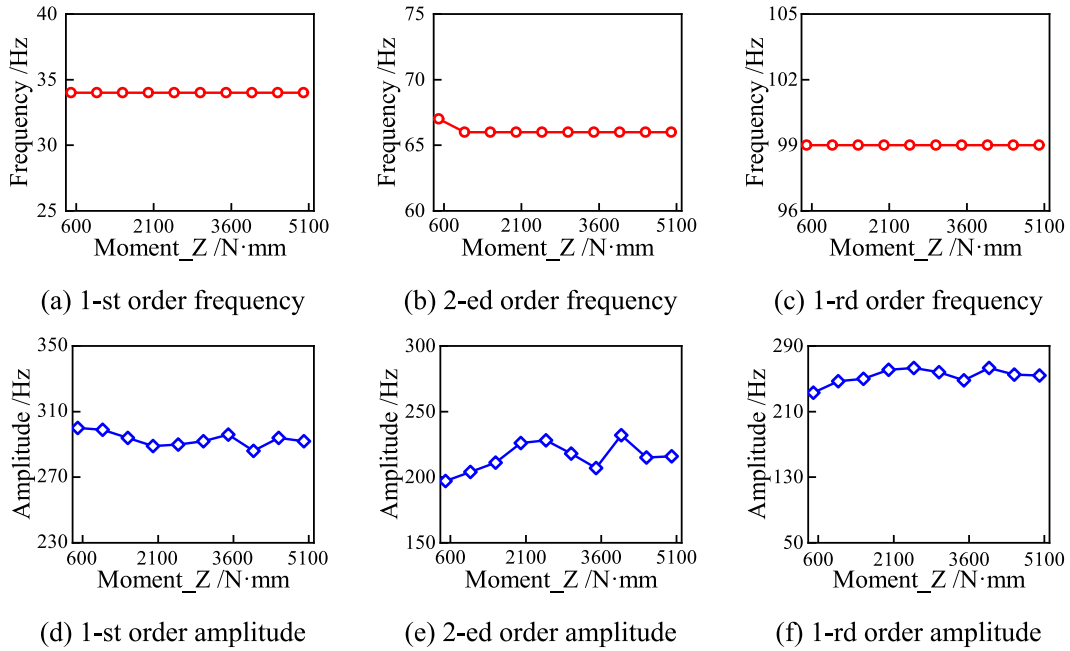


Fig. 27. Variation of frequency and amplitude for contact load on inner raceway (M_z).

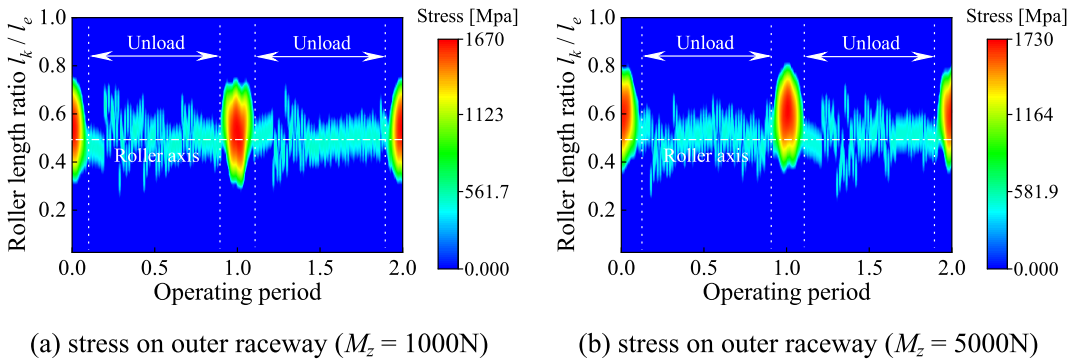


Fig. 28. Effect of z-axis moment on contact feature of outer raceway.

4.3. Effect of z-axis moment

Lastly, it is researched the effect of z-axis moment on the bearing performance. As shown in Fig. 13(c), continue to maintain a driving speed of 5000 r/min and a radial load of 3000 N, the z-axis moment is grown from N·mm to 5000 N·mm, and the dynamic performance between the roller and raceway is further simulated.

(a) contact feature.

The variation of the contact load on the inner raceway with different z-axis moments is indicated in Fig. 26. As shown in Fig. 26(a) and (b), there is little effect of the z-axis moment on the contact load in both the time and frequency domains. By comparing Fig. 26(c) with Fig. 26(d), the increase in the z-axis moment not only increases the maximum contact stress, but also causes the position to move vertically along the roller axis. In addition, as the moment increase, the size of the loaded region has not changed significantly.

In order to quantitatively analyze the effect of moments on the first three orders of frequency, Fig. 27 gives their amplitude and frequency at different moments. The effect of the z-axis moment on load frequency and amplitude is negligible except for the first-third order frequency.

As shown in Fig. 28, the effect of the z-axis moment on the outer raceway in the loaded region is similar to that on the inner raceway. On the contrary, in the unloaded region, an increase in the z-moment reduces the inclination of the roller.

(b) sliding feature.

Fig. 29 presents the effect of the z-axis moment on sliding feature of inner raceway. As shown in Fig. 29(a), the sliding velocity of the outer raceway fluctuates unevenly at a lower frequency compared to that of the y-axis moment. In the loaded region, the sliding

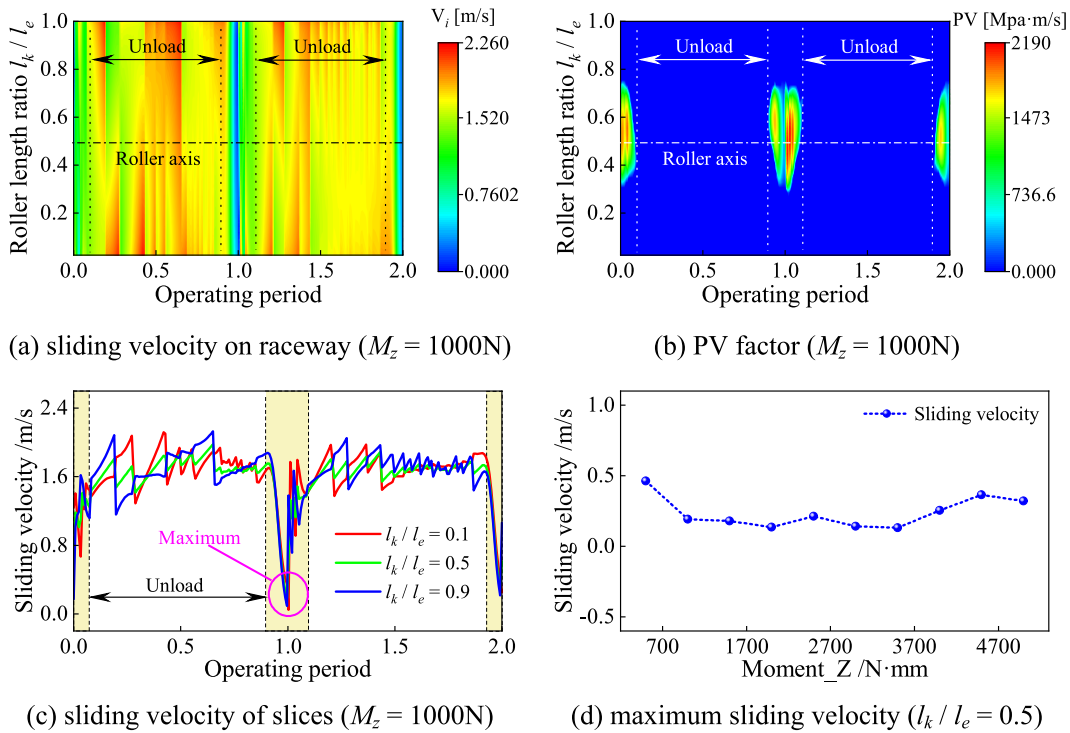


Fig. 29. Effect of z-axis moment on sliding feature of inner raceway.

velocity first decreases and then increase, and there is a maximum value exists. From the PV factor in Fig. 29(b), there are two extremes in one loaded region, with the extreme in the back half of loaded region being smaller than the front half. Fig. 29(c) further shows the variation of the sliding velocity on the typical slices. In the loaded region, the sliding velocity falls faster than it rises. In the rising phase, the sliding velocity fluctuates in a sawtooth pattern. As shown in Fig. 29(d), as the moment increases, the sliding velocity in the model slice decreases first and fluctuates in a small range.

The variations of the roller speeds under different z-axis moments are further presented in Fig. 30. The effect of the z-axis moment on roller speeds is very small and almost negligible, except for the first-third order frequency of the orbital speed.

As shown in Figs. 25 and 31, compared with the y-axis moment, the action of the z-axis moment results in a smaller change in the sliding velocity of the outer raceway. It means that an increase in the z-axis moment reduces the deflection of the roller. As far as other velocity variation characteristics are concerned, both moments have almost the same effect on the outer raceway.

To provide a clear and concise comparison of the bearing's performance under the three conditions, Table 4 summarizes the prediction results. It shows that increasing the radial load significantly affects the contact and sliding characteristics inside the bearing. As the radial load increases, both contact stress and load rise, while the inner raceway's sliding speed decreases, indicating better control of bearing sliding. As for the moment about the y-axis, an increase in moment does not significantly alter the stress due to the enlarged load zone. The moment about the z-axis has a negligible effect on performance parameters. However, as Figs. 20, 22, 26 and 28 shown, while increasing the moment may not change the value of contact indicators, it does affect the shape of the loaded zone on the raceway.

5. Conclusion

A sliding method considering roller elastic deformation is first proposed. A dynamic model of cylindrical roller bearings is further constructed and the effects of different loading methods on the dynamic performance inside the bearing are further investigated. Some interesting conclusions are drawn.

- (1) Due to the consideration of elastic deformation, the results predicted by the slicing method based on beam is the closest to that of the FE model compared to other common methods.
- (2) A rich spectrum of frequencies emerges in the bearing performance index because of the roller deformation. These frequencies are all determined by the roller passing frequency.
- (3) When sliding occurs inside the bearing, the sliding velocity is determined by the orbital speed of the roller, so a high orbital speed is required to minimize the effects of sliding.

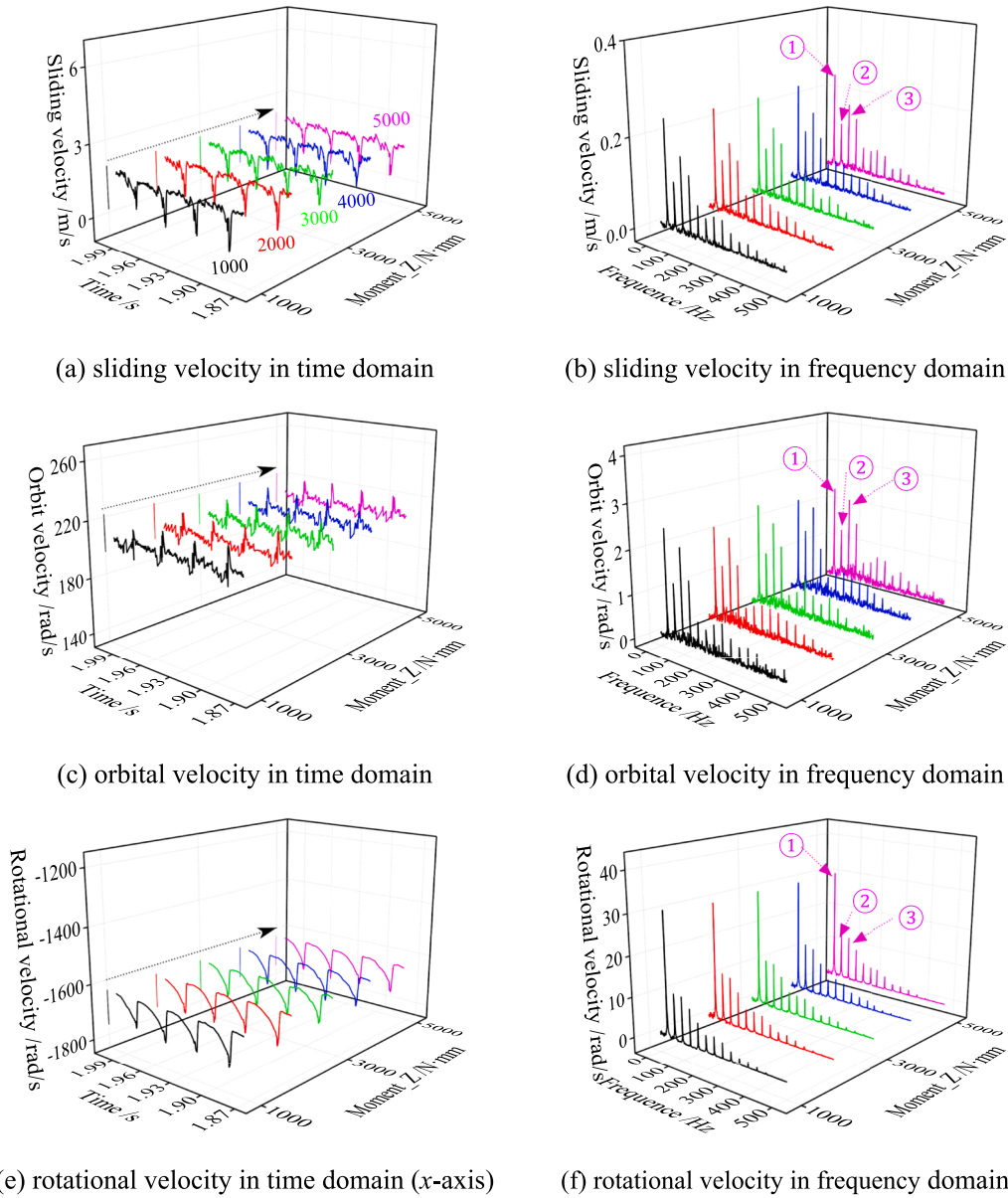


Fig. 30. Relationship between sliding velocity and rotation speed (M_z).

- (4) The y-axis moment helps to widen the loaded zone, reduce maximum contact stresses, and extend the bearing fatigue life. However, excessive moment can increase the loaded zone from one to two.

CRediT authorship contribution statement

Ke Yan: Writing – review & editing, Validation, Software. **Shuaijun Ma:** Writing – original draft, Validation, Software, Investigation. **Bin Fang:** Writing – review & editing, Software, Investigation. **Fei Chen:** Writing – review & editing, Validation, Resources. **Jun Hong:** Writing – review & editing, Validation, Project administration. **Pan Zhang:** Validation, Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

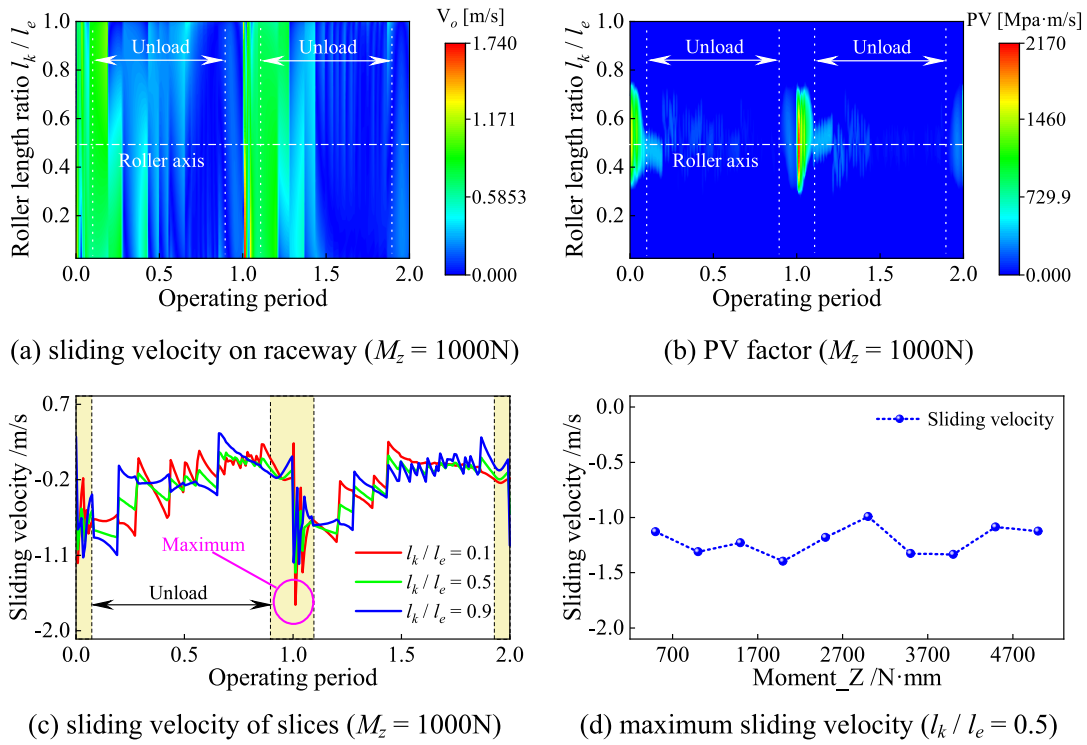


Fig. 31. Effect of z-axis moment on sliding feature of outer raceway.

Table 4

Comparisons of performance features under different loading conditions.

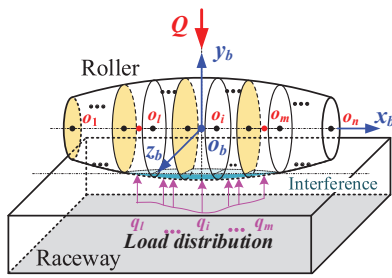
Indicator		Loading condition		
		y-axis load	y-axis moment	z-axis moment
		y-axis load	y-axis moment	z-axis moment
Contact	Load	Increasing	Decreasing	Almost unchanged
Sliding	Stress	Increasing	Almost unchanged	Almost unchanged
	Sliding velocity(Inner)	Decreasing	Almost unchanged	Almost unchanged
	Sliding velocity(Outer)	Almost unchanged	Almost unchanged	Almost unchanged
	Orbital velocity	Increasing	Almost unchanged	Almost unchanged
	Rotational velocity	Increasing	Almost unchanged	Almost unchanged

Acknowledgement

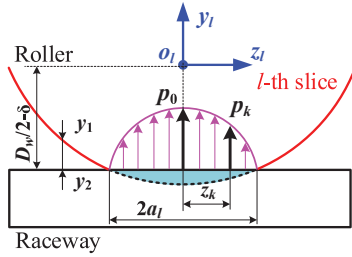
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Appendix A. . Slicing method of Hertzian contact

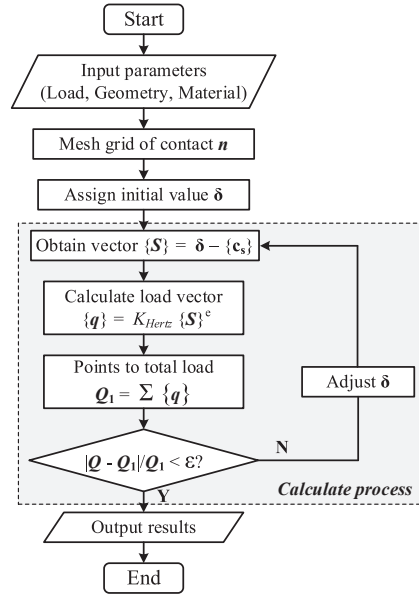
The slicing method based on Hertzian contact is presented. Its detailed solution flow is presented in Fig. A1. As shown in Fig. A1(a), the roller is divided into n pieces along its axis. The sum of these loads on all the slices is equal to the external load. Therefore, the assigned loads for these slices should first be obtained.



(a) roller discrete schematic



(b) individual slice stress



(c) solution flowchart

Fig. A1. Solution for roller stress distribution based on Hertzian contact theory

With l -th slice, for example, as shown in Fig. A1(b), its normal load can be obtained by the following equation.

$$q_l = K_{Hertz} \cdot \left\{ (D_w/2 - \delta - C_s^l \cdot 0)_{\max} \right\}^{10/9} \quad (\text{A-1})$$

where D_w is the diameter of the roller. δ is the displacement of the roller center. C_s^l is the trimming factor for l -th slice, and for logarithmic trimming, it can be expressed as:

$$C_s^l = A \cdot \ln \left[1 - (2x_l/L)^2 \right]^{-1} \quad (\text{A-2})$$

where A is the factor of logarithmic trimming. L is the length of the roller.

K_{Hertz} denotes Hertzian contact deformation coefficient of the l -th slice, defined as [24]:

$$K_{Hertz} = \pi \cdot (L/n)^{8/9} E' / 3.891^{10/9} \quad (\text{A-3})$$

where E' is the material-related coefficient, $E' = 1/[(1-\nu_1^2)/E_1 + (1-\nu_2^2)/E_2]$. ν is the Poisson's ratio. E is the Young's modulus.

The equilibrium equation for the whole roller can be further derived as:

$$Q_{External} = \sum_{l=1}^n q_l = \sum_{l=1}^n K_{Hertz} \cdot (\max(D_w/2 - \delta - C_s^l \cdot 0))^{10/9} \quad (\text{A-4})$$

where $Q_{External}$ is the external load applied on roller.

The essence of this slicing method of Hertzian contact lies in solving Eq. (A-4). The solution flow of this method is further given in Fig. A1(c). For simplicity, the performance metrics in the flowchart are represented as vectors.

Once the displacement of the roller is known, the load distribution along the roller axis can be obtained. With the help of Hertzian line contact theory[24], the stress distribution in the roller/raceway interface can be further calculated by the following equation.

$$p_k = p_0 \sqrt{1 - (z_k/a_l)^2} \quad (\text{A-5})$$

where p_0 and a_l are the maximum stress and contact half-width, respectively, defined as:

$$\begin{cases} p_0 = \sqrt{E' q_l / (\pi R)} \\ a_l = \sqrt{4R q_l / (\pi E')} \end{cases} \quad (\text{A-6})$$

where R is equivalent radius of curvature.

Appendix B. . Slicing method of influence coefficient

The slicing method based on influence coefficient is introduced in Appendix B, which considers the interactions between the slices. Similar to the slicing method of Hertzian contact, its solution is displayed in Fig. B1.

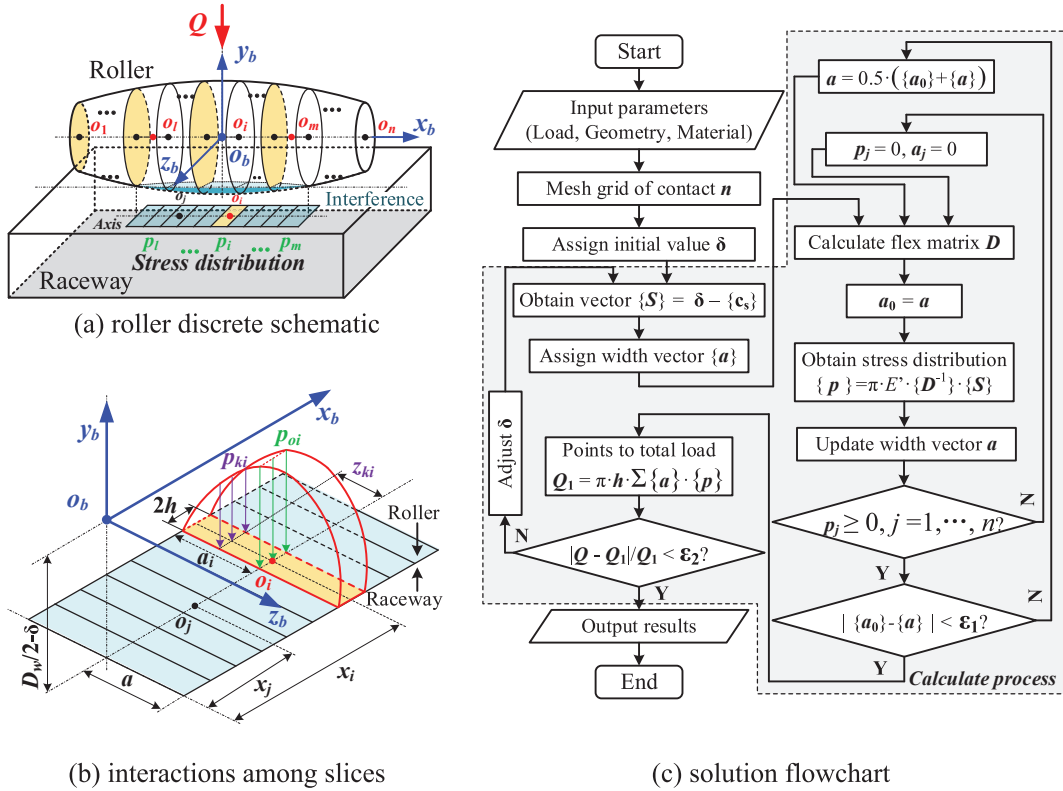


Fig. B1. Solution for roller stress distribution based on influence coefficient method (Non-Hertz)

As shown in Fig. B1(a), the roller is cut into n slices along its axis. However, only m of these slices are in contact with the raceway. The interactions among them and their stress distributions are shown in Fig. B1(b). In addition to its own deformation, the j -th slice produces a small but non-negligible displacement at the center of the i -th slice, defined as:

$$w_{ij} = p_{0j} D_{ij} / (\pi E') \quad (\text{B-1})$$

where E' is the material-related coefficient, $E' = 1/[(1-\nu_1^2)/E_1 + (1-\nu_2^2)/E_2]$. E is the Young's modulus. ν is the Poisson's ratio. p_{0j} is the maximum stress of the j -th slice. D_{ij} is the flexibility coefficient, defined by [29]:

$$D_{ij} = \int_{-a_j}^{a_j} \int_{z_j-h}^{z_j+h} \frac{\sqrt{1 - (x'/a_j)^2}}{\sqrt{x'^2 + (z_i - z_j - z')^2}} dx' dz' \quad (\text{B-2})$$

where h is half the width of the slice, $h = L/(2n)$. a_j is the contact width of the j -th slice, which can be obtained by:

$$a_j = 2R \cdot p_{0j} / E' \quad (\text{B-3})$$

For the contact between the roller and raceway, the flexibility coefficient can be further expressed as:

$$D_{ij} = \int_{-a_j}^{a_j} \sqrt{1 - (x'/a_j)^2} \ln \left(\frac{|z_i - z_j| + h + \sqrt{x'^2 + (z_i - z_j + h)^2}}{|z_i - z_j| - h + \sqrt{x'^2 + (z_i - z_j - h)^2}} \right) dx' \quad (\text{B-4})$$

Therefore, the equilibrium equation of the roller and individual slices is derived as:

$$\pi h \sum_{j=1}^n a_j p_{0j} = Q_{\text{External}} \quad (\text{B-5})$$

$$\frac{1}{\pi E'} \sum_{j=1}^n D_{ij} p_{0j} = \delta - C_s^j \quad (\text{B-6})$$

where Q_{External} is the external load applied on roller. δ is the displacement of the roller center. C_s^j is the trimming factor for j -th slice, and for logarithmic trimming, it can be expressed as:

$$C_s^j = A \cdot \ln \left[1 - (2x_j/L)^2 \right]^{-1} \quad (\text{B-7})$$

where A is the factor of logarithmic trimming. L is the length of the roller.

The essence of this method is to solve Eqs. (B-5)-(B-6) to obtain the stresses on each slice (p_{0j}) and the roller displacement (δ). Fig. B1(c) further presents the solution flowchart of these equations detail. It is worth noting that the slices with negative stresses need to be eliminated for each iteration since they are not in contact with the raceway.

Data availability

No data was used for the research described in the article.

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